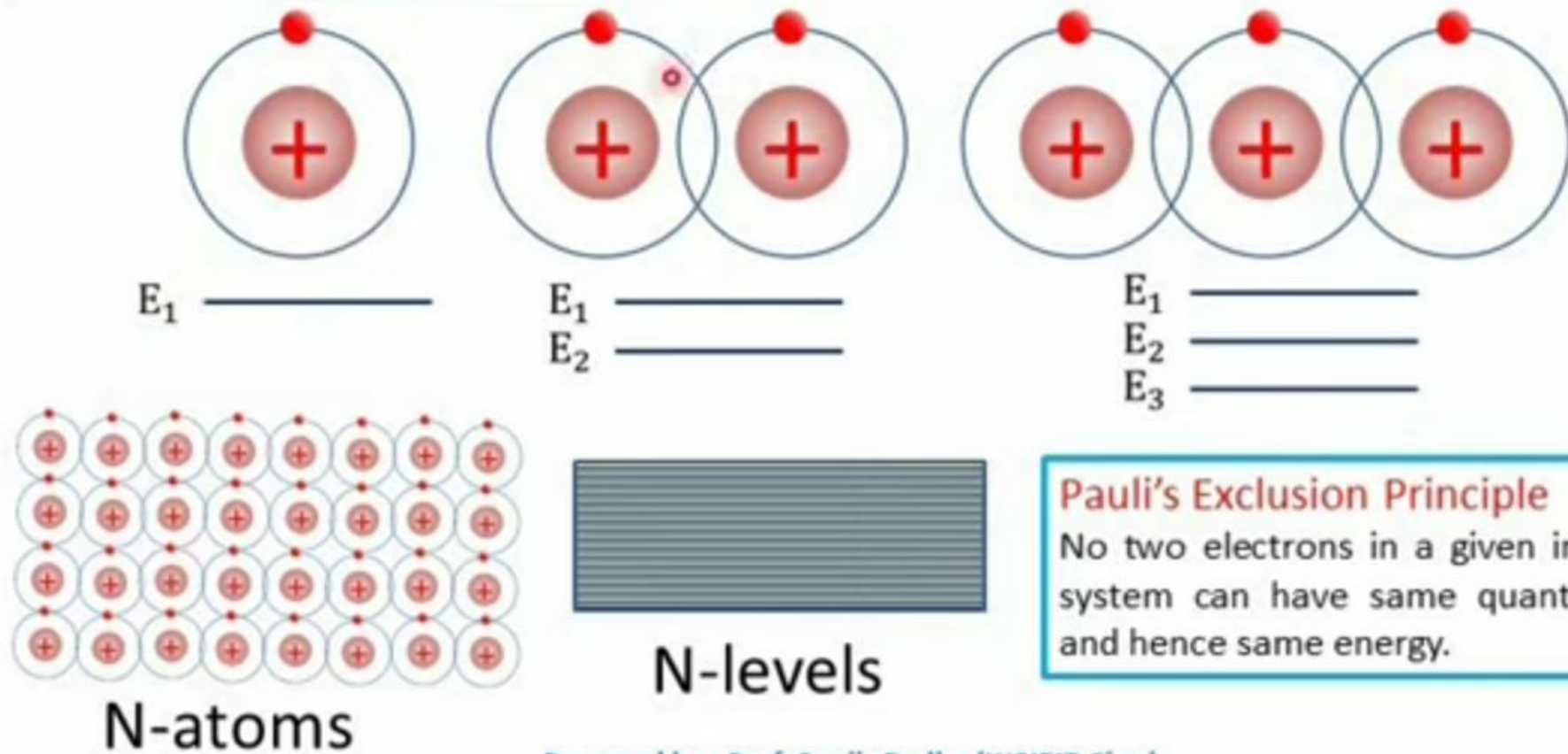


Module 3: Semiconductor Physics

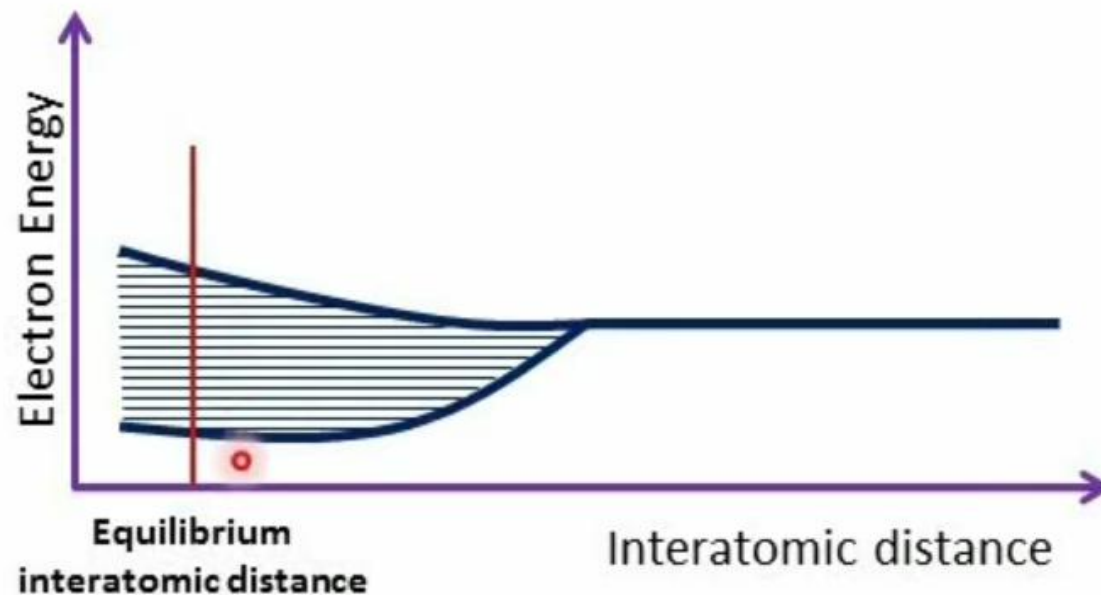
Formation of Energy Bands



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Module 3: Semiconductor Physics

Formation of Energy Bands

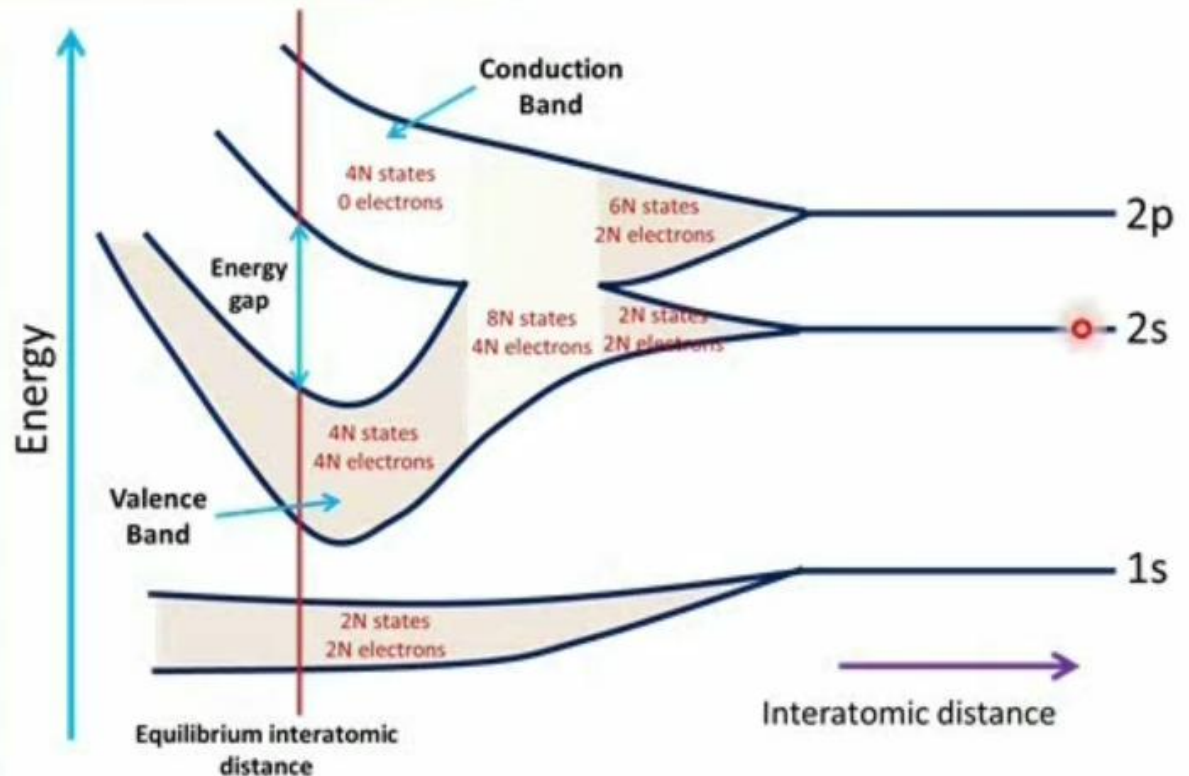


Pushing the atoms together, the initial quantized energy level will split into a band of discrete energy levels.

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Formation of Energy Bands in Diamond

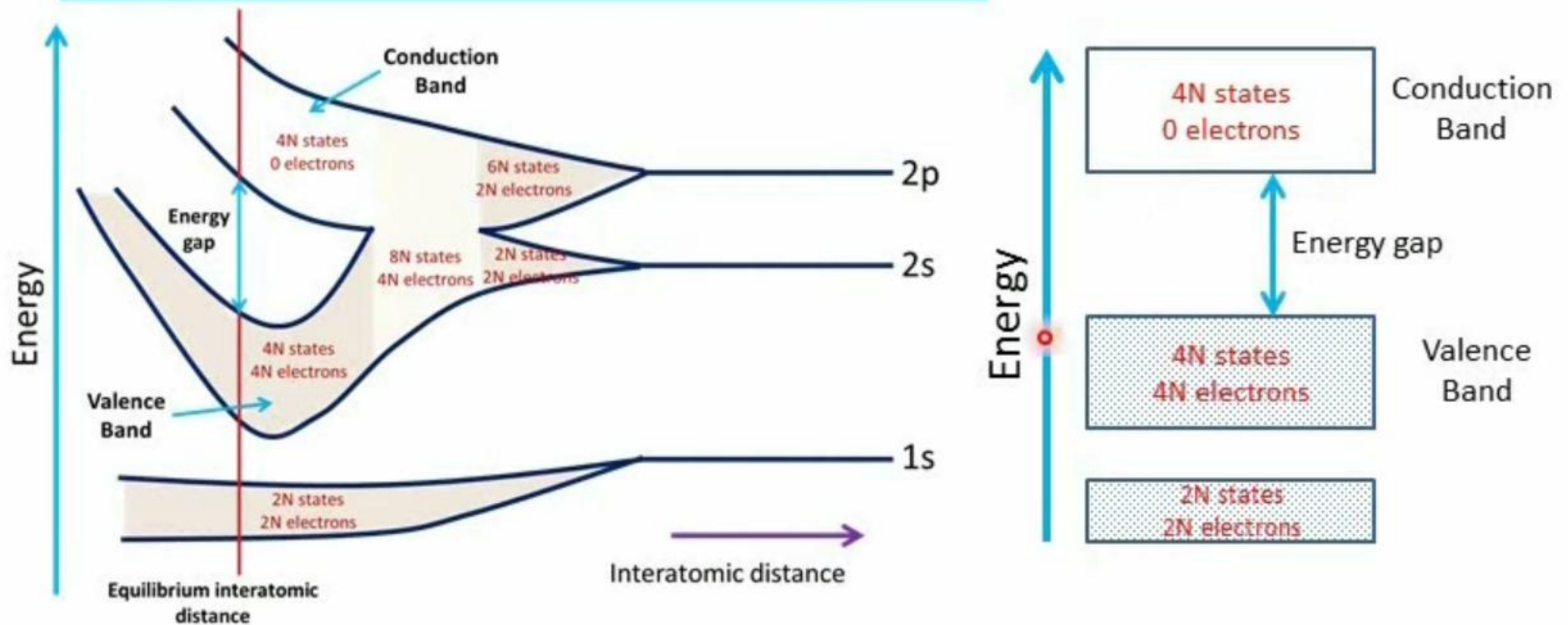
- Electronic structure individual carbon atom:
 $1s^2 2s^2 2p^2$
- Each atom has two 1s states, two 2s states, six 2p states and the higher states.
- Hence for N atoms, there will be $2N$, $2N$ and $6N$ available states of type 1s, 2s, and 2p respectively.



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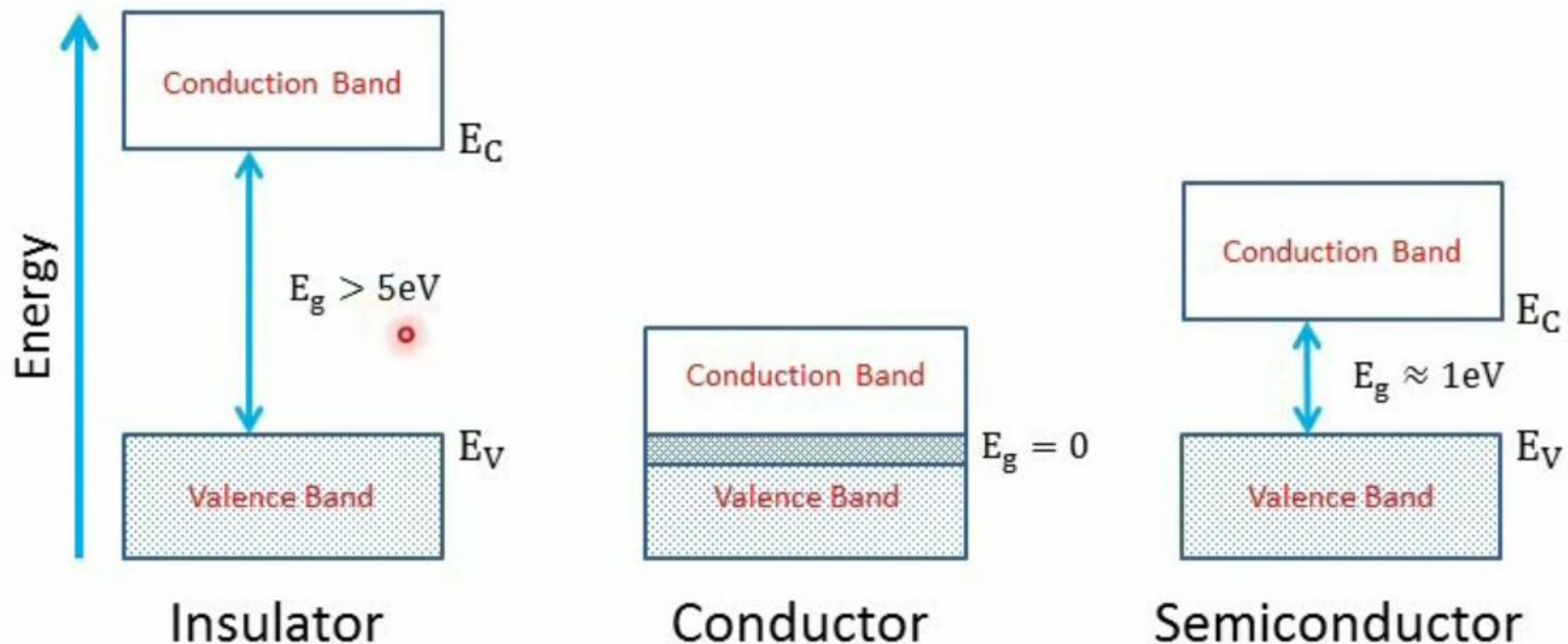
Formation of Energy Bands in Diamond



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Module 3: Semiconductor Physics

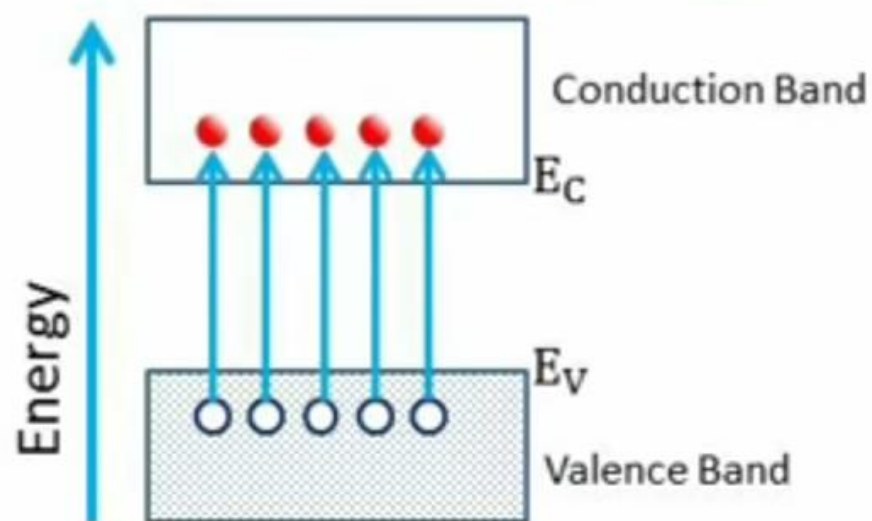
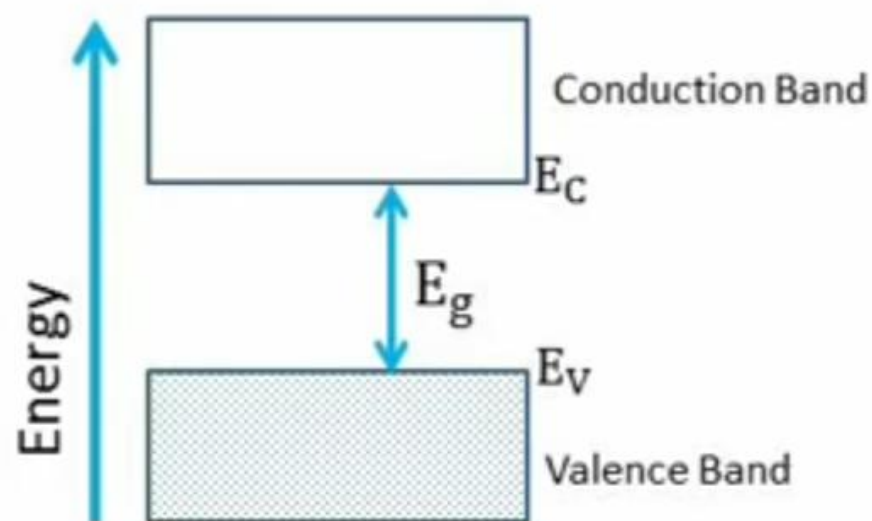
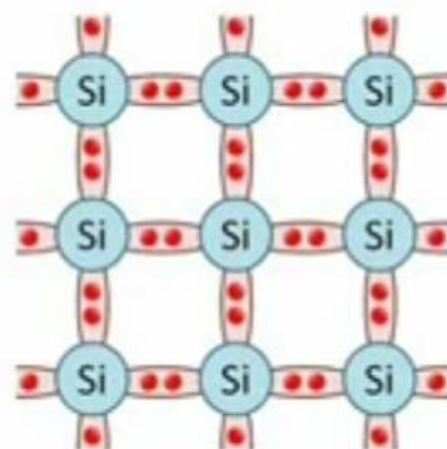
Classification of solids on the basis of band theory



Prepared by : Prof. Sanjiv Badhe (KJSIEIT, Sion)

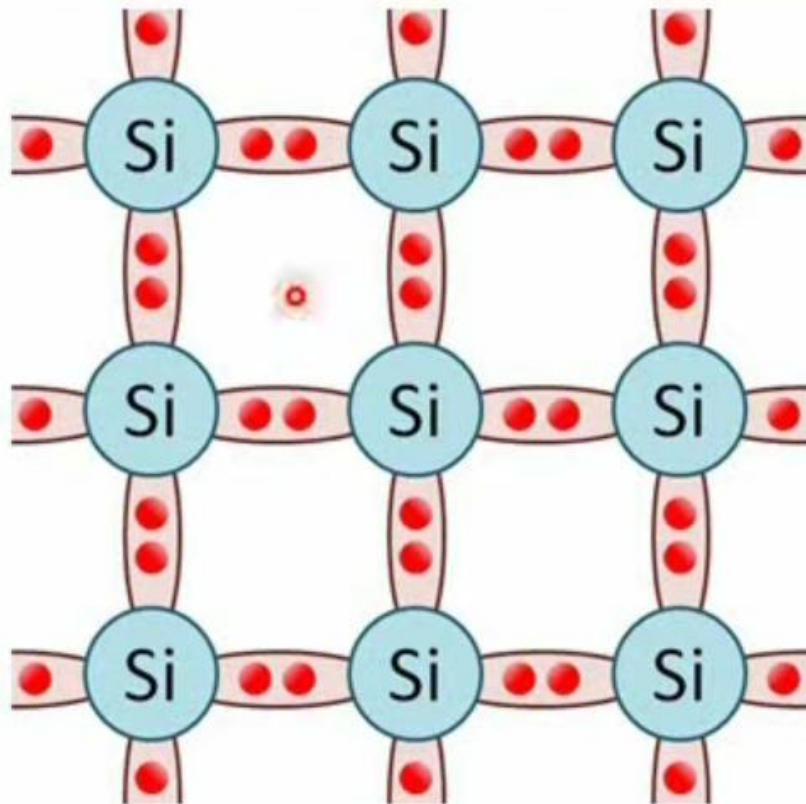
Module 3: Semiconductor Physics

Intrinsic Semiconductors



Module 3: Semiconductor Physics

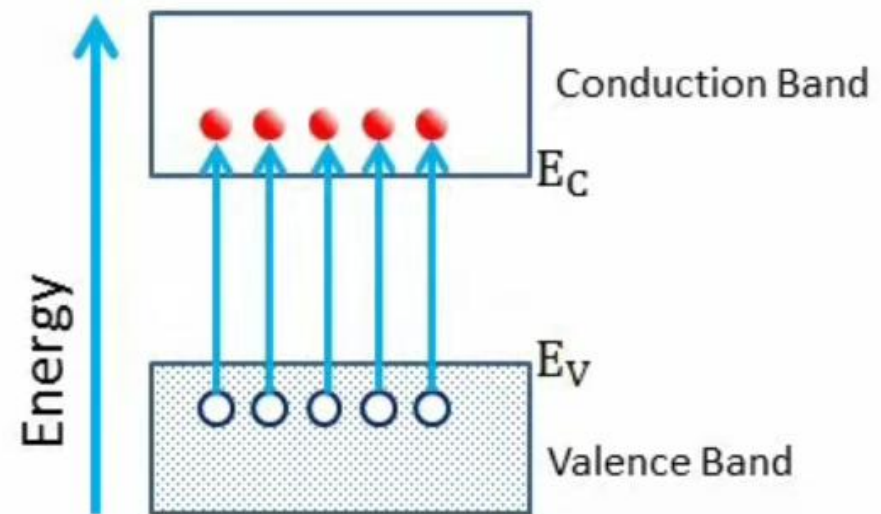
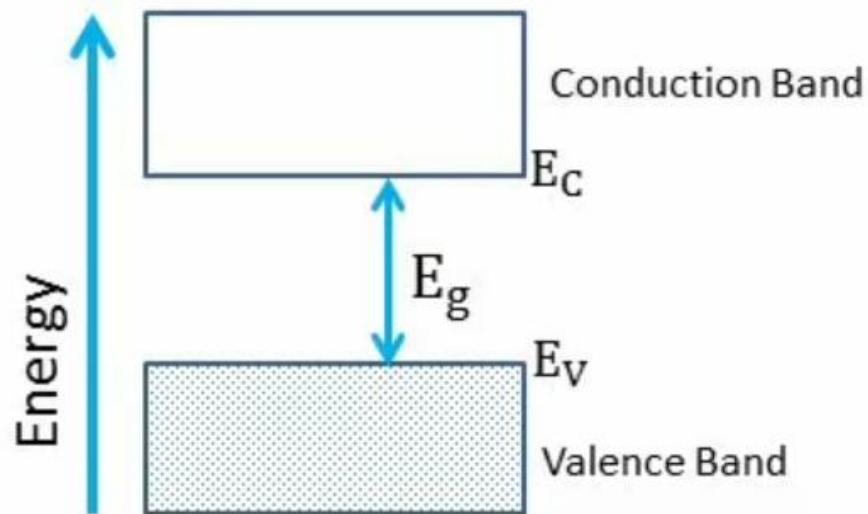
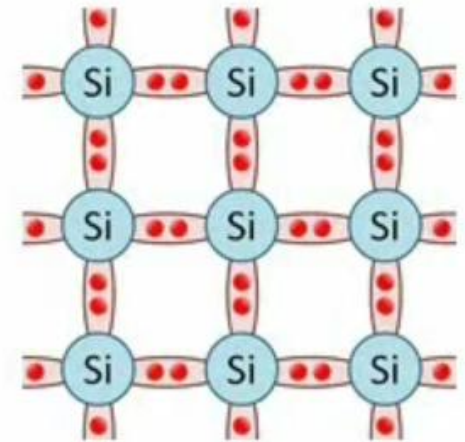
Covalent Bonds in Semiconductors



Neighboring atoms form covalent bonds by sharing electrons with each other

Module 3: Semiconductor Physics

Intrinsic Semiconductors



Module 3: Semiconductor Physics

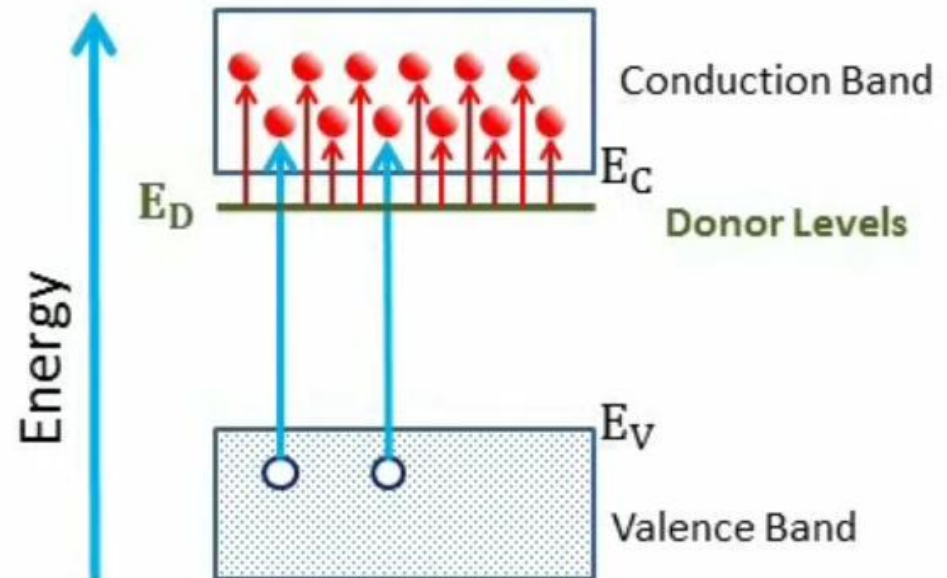
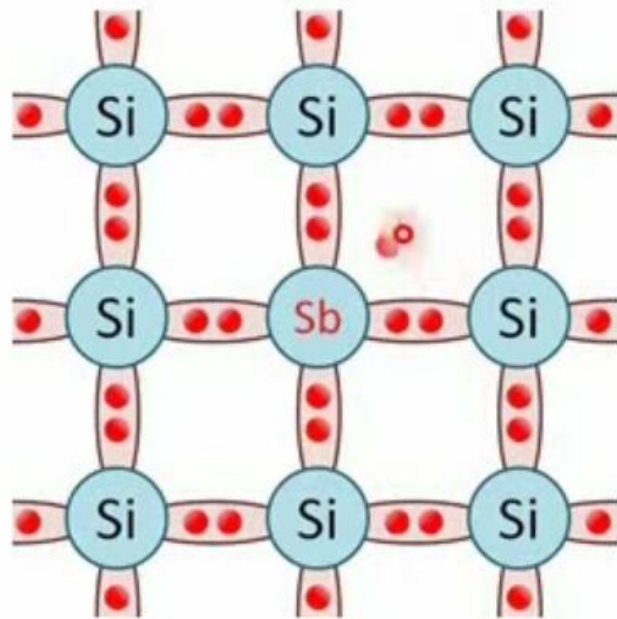
Why Silicon is preferred over Germanium?

- Silicon and Germanium are two most widely used semiconductors.
- Silicon is preferred over Germanium because of its **superior thermal properties**.
- The maximum operational temperature for Ge-devices is 80°C while Si-devices may be used at up to 160°C .
- Also, at room temperature, **silicon crystal has almost no free electrons compared with Germanium crystal**

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Extrinsic Semiconductors (Doped Semiconductors)

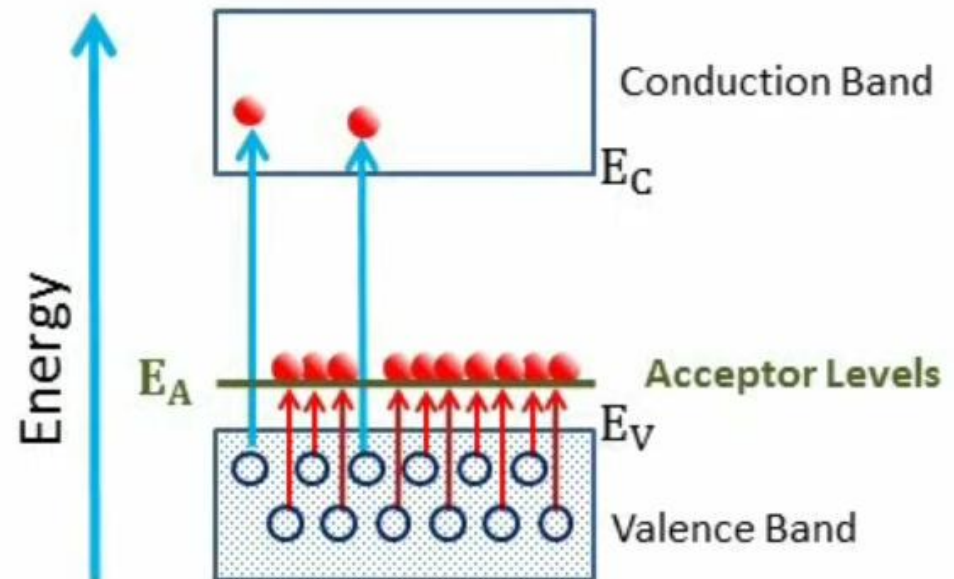
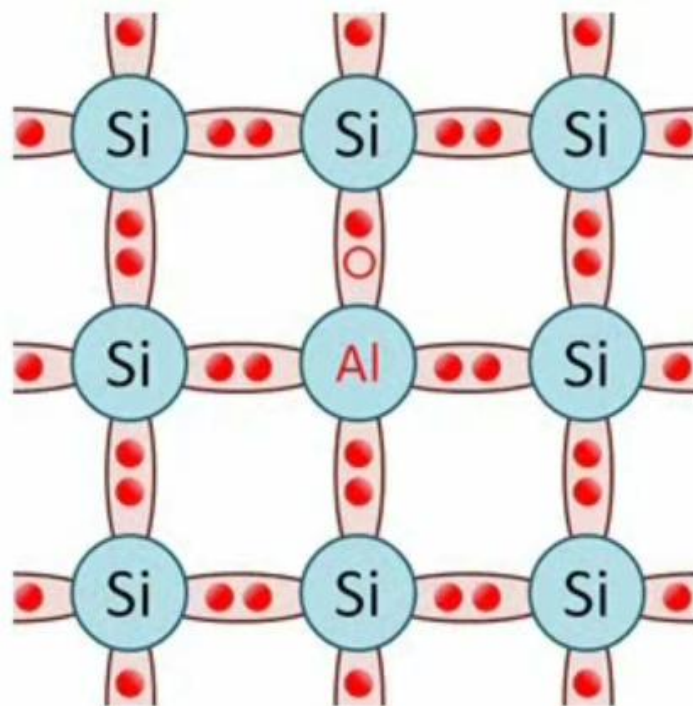
n-type semiconductor



Module 3: Semiconductor Physics

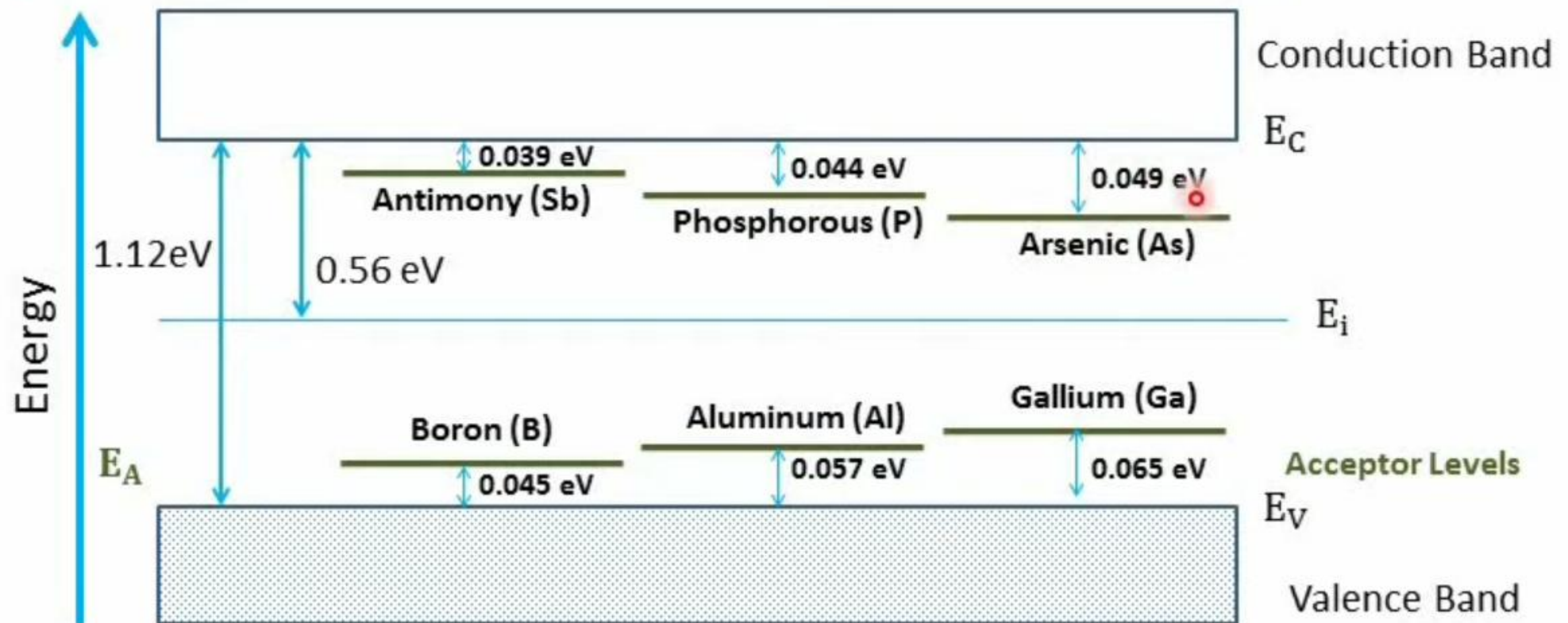
Extrinsic Semiconductors (Doped Semiconductors)

p-type semiconductor



Module 3: Semiconductor Physics

Extrinsic Semiconductors (Doped Semiconductors)



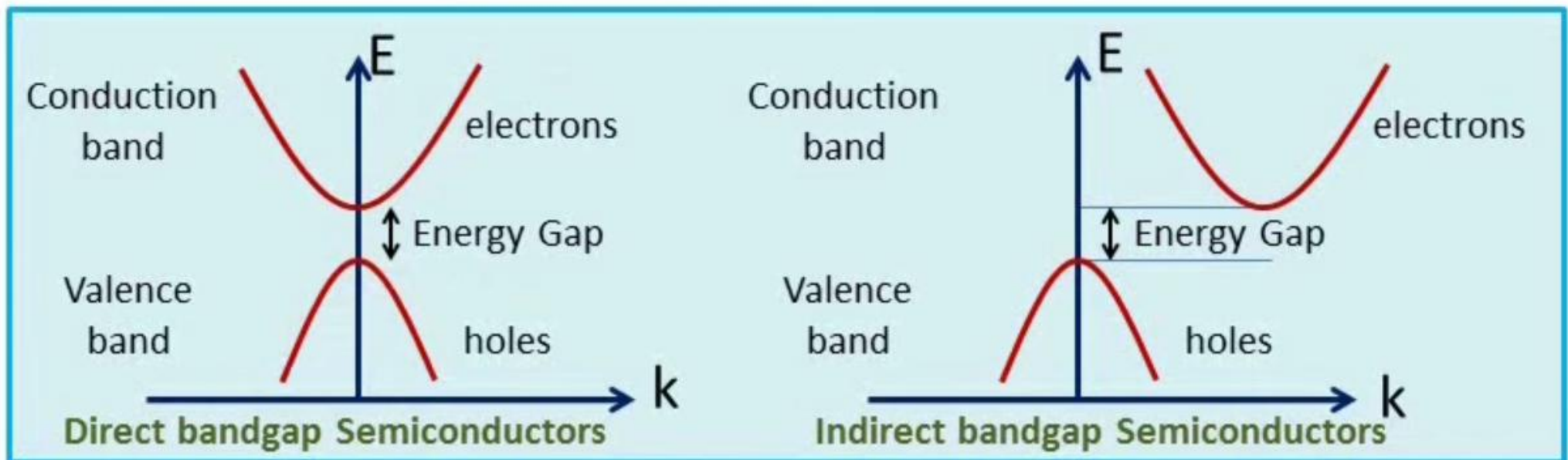
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Direct Gap and Indirect Gap Semiconductors

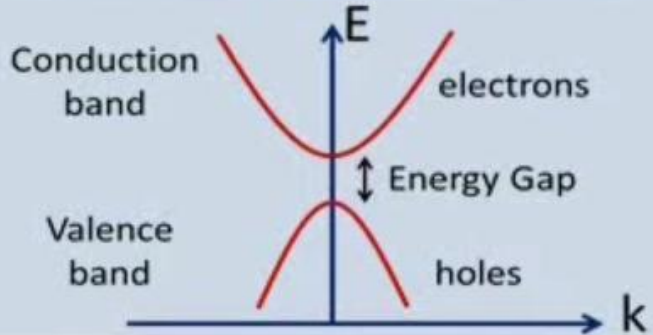
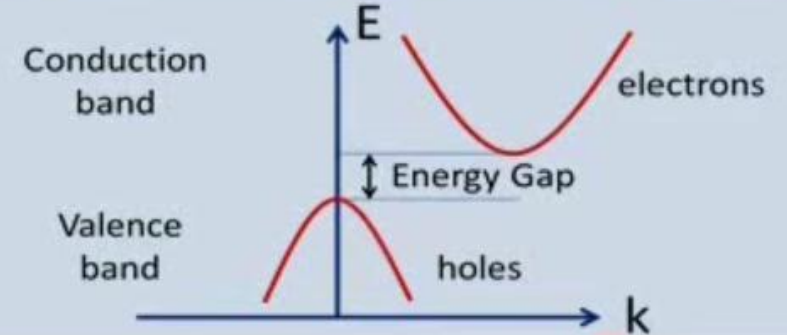
The energy of an electron is given by $E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$ where p is momentum, m is mass of an electron, $\hbar$ is Planck's constant and k is propagation constant

Thus $E \propto k^2$ which is an equation of parabola. The graph of E vs k is shown below -



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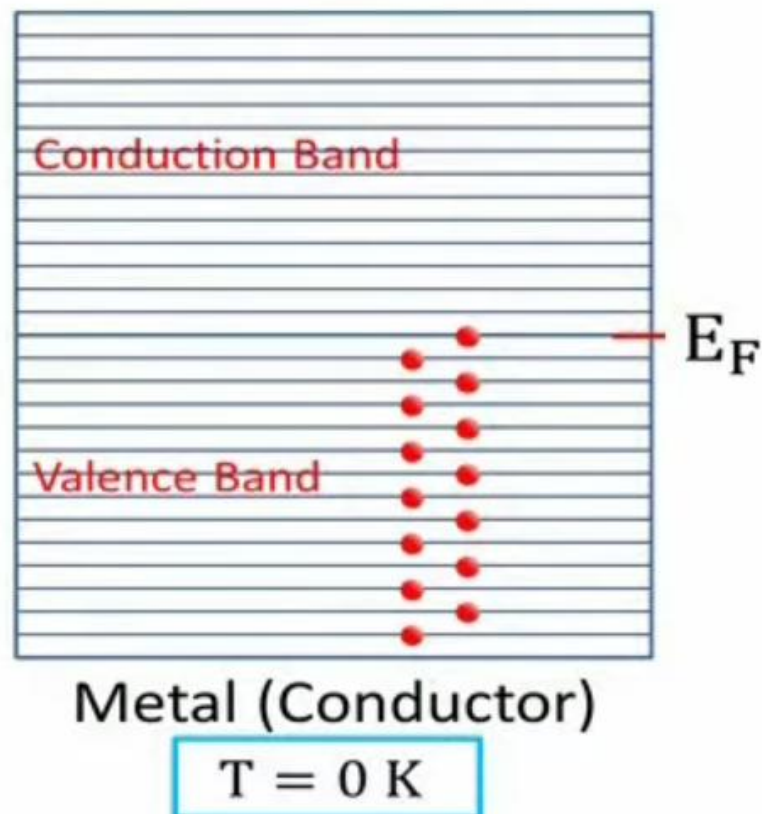
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Direct bandgap Semiconductors	Indirect bandgap Semiconductors
	
Maximum of valence band and minimum of conduction band occur at same momentum values	Maximum of valence band and minimum of conduction band occur at two different momentum values.
Electron making a transition from valence band to conduction band need not undergo any change in its momentum.	In order to make a transition from maximum point in valence band to minimum point in conduction band, the electron requires energy for the change in momentum in addition to the energy gap E_g
The compound semiconductors such as GaAs, are direct gap semiconductors	All elemental semiconductors such as Si, Ge, are indirect gap semiconductors
These direct gap semiconductors are used in LED and Semiconductor Lasers.	Not useful for LEDs and Semiconductor Lasers

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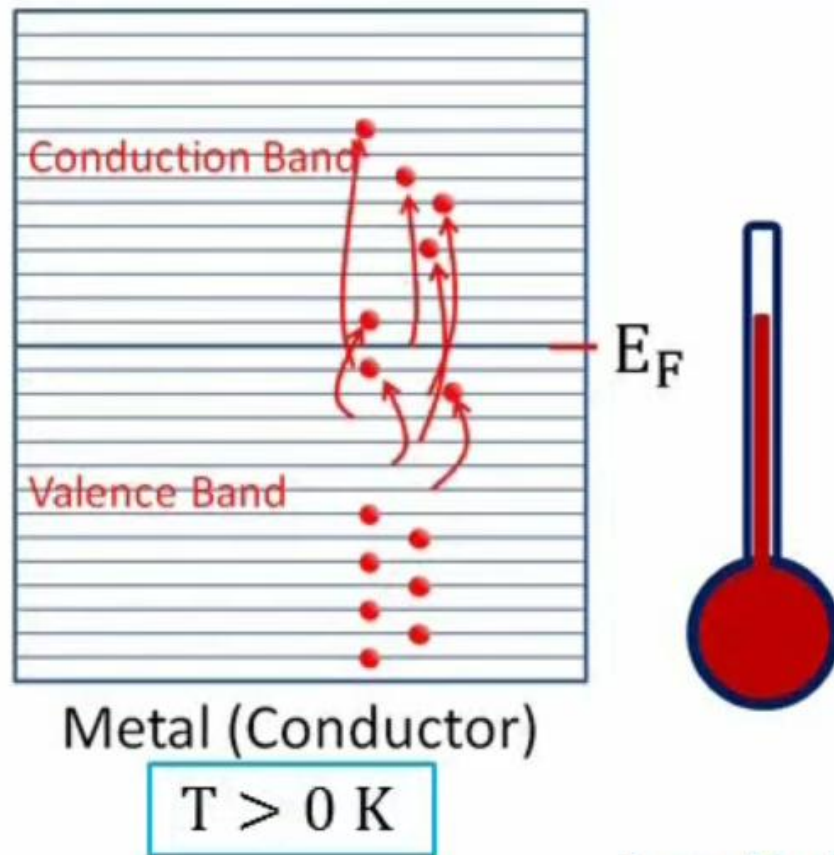
Fermi Level



- In metal, there is one partially filled band which is a result of conduction band overlapping with valence band.
- In this band lowest energy levels are filled first.
- The highest occupied energy level at absolute zero temperature (0K) is called as Fermi level.
- Energy corresponding to it is called as **Fermi energy denoted by E_F**

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Fermi Level and Fermi-Dirac Distribution Function



- At temperature $T > 0\text{ K}$, the distribution of electrons over a range of allowed energy levels at thermal equilibrium is given by Fermi-Dirac Distribution function

$$f(E) = \frac{1}{1 + e^{\left(\frac{E - E_F}{kT}\right)}}$$

$f(E)$ is the probability of occupancy for energy level E

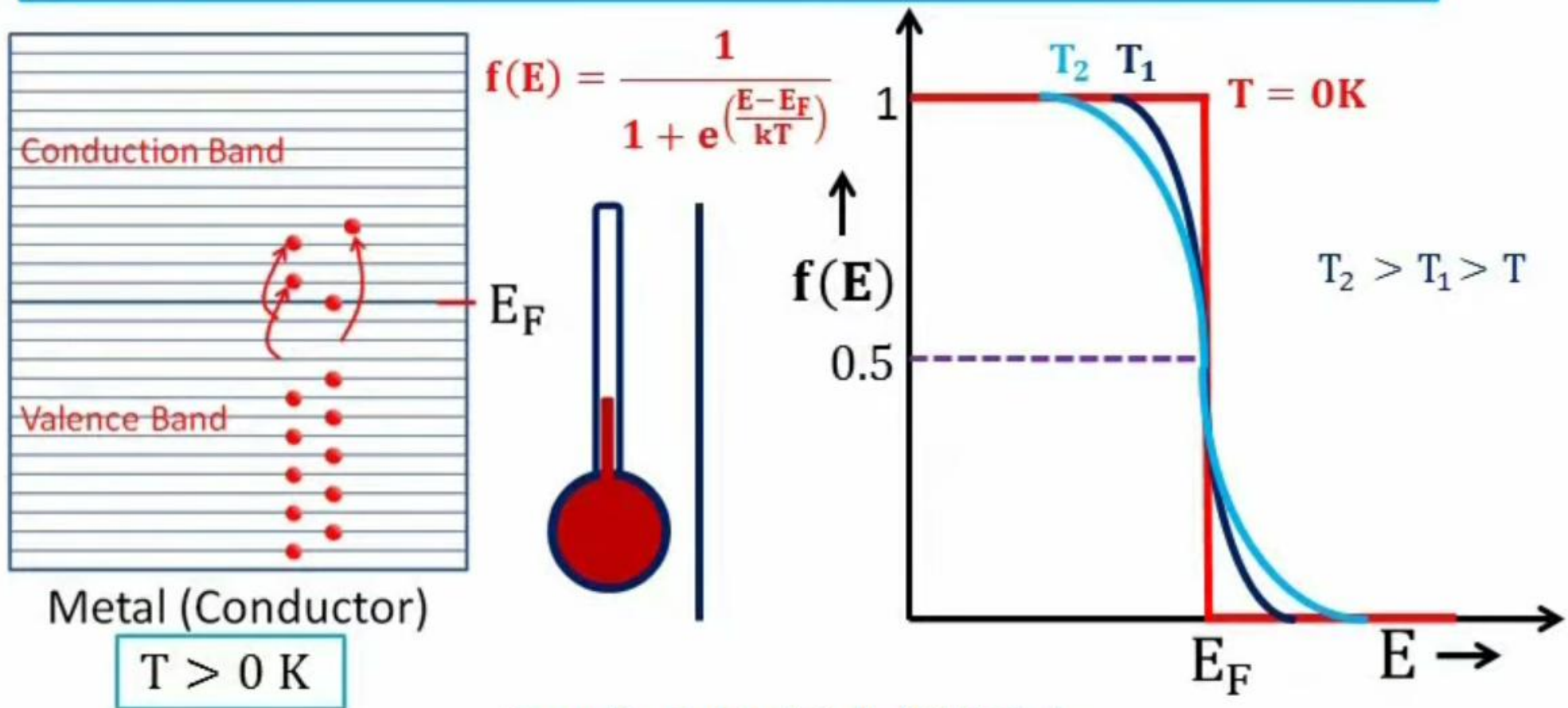
E_F is Fermi energy

T is temperature in $^{\circ}\text{K}$ and

$k = 1.38 \times 10^{-23} \text{ J/K} = 8.625 \times 10^{-5} \text{ eV/K}$

Module 3: Semiconductor Physics

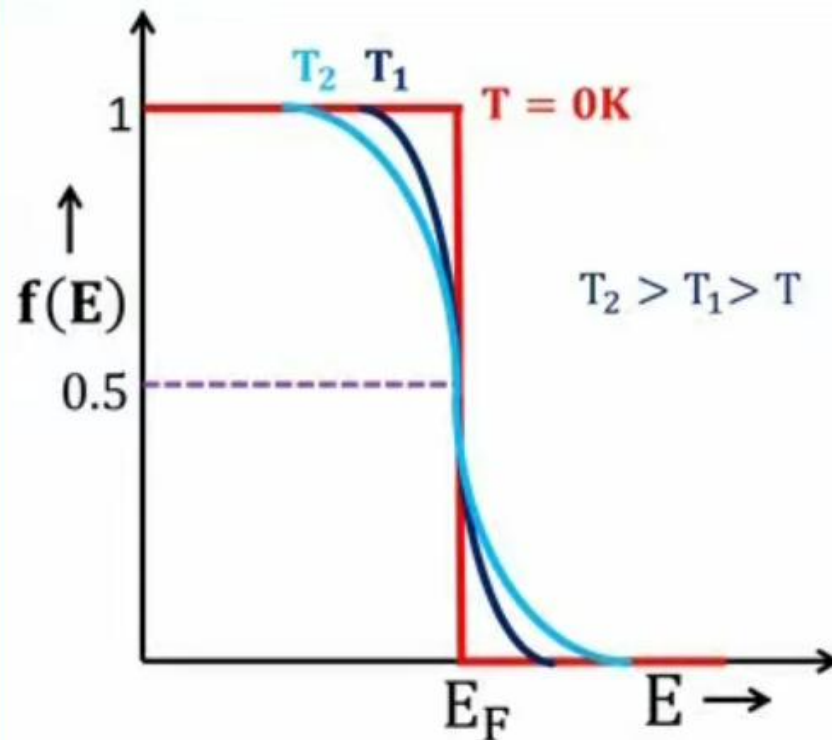
Fermi-Dirac Distribution Function at various temperatures



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Fermi-Dirac Distribution Function at various temperatures



$$f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}}$$

At $T = 0 K$, For $E < E_F$

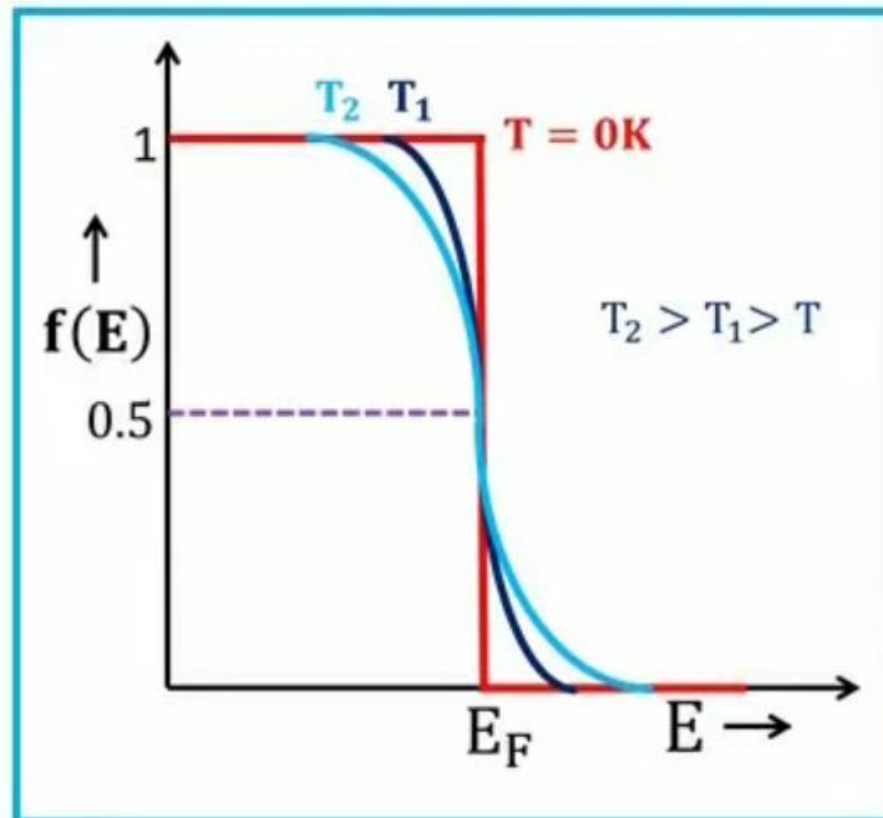
$$e^{\left(\frac{E-E_F}{kT}\right)} = e^{\frac{-ve\ number}{0}} = e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

$$\therefore f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}} = \frac{1}{1 + 0} = 1$$

This indicates all energy levels below E_F are completely filled at absolute zero temperature

Module 3: Semiconductor Physics

Fermi-Dirac Distribution Function at various temperatures



$$f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}}$$

At $T = 0 K$, For $E > E_F$

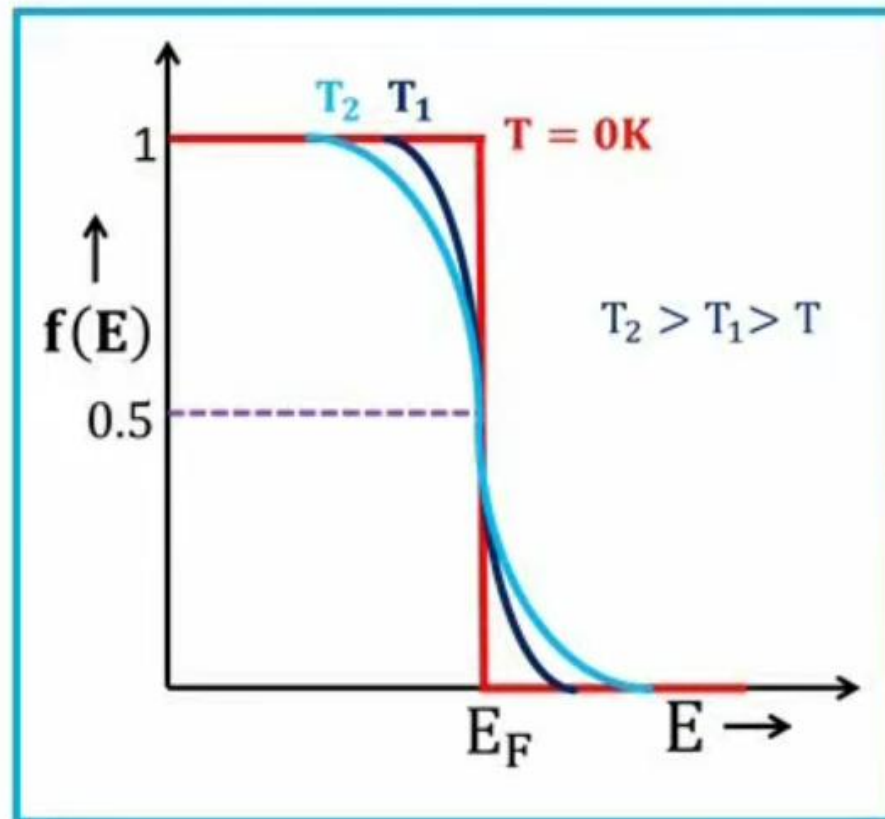
$$e^{\left(\frac{E-E_F}{kT}\right)} = e^{\frac{+ve \text{ number}}{0}} = e^{+\infty} = \infty$$

$$\therefore f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}} = \frac{1}{1 + \infty} = 0$$

This indicates all energy levels above E_F are completely empty at absolute zero temperature

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Fermi-Dirac Distribution Function at various temperatures



$$f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}}$$

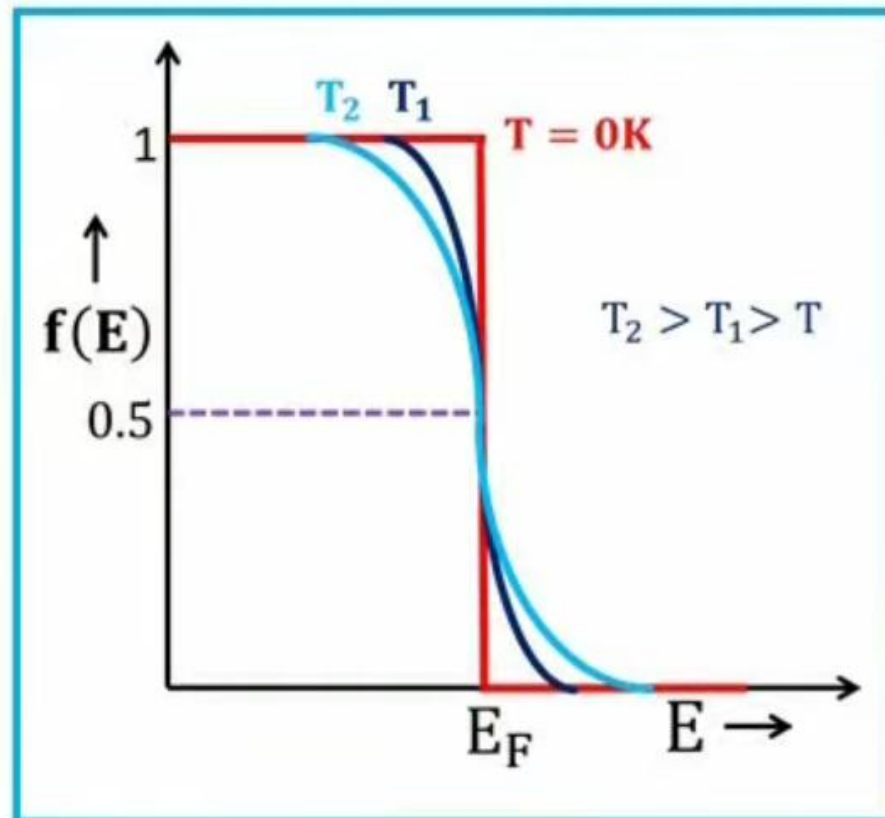
At $T = 0\text{ K}$, For $E = E_F$

$$e^{\left(\frac{E-E_F}{kT}\right)} = e^{\frac{0}{0}} = \text{NOT DEFINED}$$

Fermi-Dirac Distribution function can not give us probability of occupancy of Fermi level at 0 K.

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Fermi-Dirac Distribution Function at various temperatures



$$f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}}$$

At $T > 0 K$, For $E = E_F$

$$e^{\left(\frac{E-E_F}{kT}\right)} = e^{\frac{0}{kT}} = e^0 = 1$$

$$\therefore f(E) = \frac{1}{1 + e^{\left(\frac{E-E_F}{kT}\right)}} = \frac{1}{1 + 1} = \frac{1}{2}$$

At $T > 0 K$, probability of occupancy for Fermi level is always half

Fermi level for silver is 5.5 eV. Find out the energy for which the probability of occupancy at 300 K is 0.9.

$$\text{Given : } T = 27^\circ\text{C} = 300^\circ\text{K}, \quad k = 8.625 \times 10^{-5} \frac{\text{eV}}{^\circ\text{K}}, \quad E_F = 5.5 \text{ eV}$$

when $f(E) = 0.9$, $E = ?$

$$\text{Probability of occupancy } f(E) = \frac{1}{1 + e^{\left(\frac{E - E_F}{kT}\right)}}$$

$$\therefore E - E_F = kT \ln \left(\frac{1}{f(E)} - 1 \right)$$

$$\therefore 1 + e^{\left(\frac{E - E_F}{kT}\right)} = \frac{1}{f(E)}$$

$$\therefore e^{\left(\frac{E - E_F}{kT}\right)} = \frac{1}{f(E)} - 1$$

$$\therefore \frac{E - E_F}{kT} = \ln \left(\frac{1}{f(E)} - 1 \right)$$

$$\therefore E = E_F + kT \ln \left(\frac{1}{f(E)} - 1 \right)$$

For $f(E) = 0.9$

$$\begin{aligned} E &= 5.5 + 8.625 \times 10^{-5} \times 300 \ln \left(\frac{1}{0.9} - 1 \right) \\ &= 5.443 \text{ eV} \end{aligned}$$

Fermi level in potassium is 2.1 eV. What are the energies for which the probability of occupancy at 300 K are 0.99 and 0.01.

Given : $T = 27^\circ\text{C} = 300^\circ\text{K}$, $k = 8.625 \times 10^{-5} \frac{\text{eV}}{^\circ\text{K}}$, $E_F = 2.1 \text{ eV}$

when $f(E) = 0.99$ and $f(E) = 0.01$, $E = ?$

Probability of occupancy $f(E) = \frac{1}{1 + e^{\left(\frac{E - E_F}{kT}\right)}}$ $\therefore E = E_F + kT \ln \left(\frac{1}{f(E)} - 1 \right)$

$$\therefore 1 + e^{\left(\frac{E - E_F}{kT}\right)} = \frac{1}{f(E)}$$

$$\therefore e^{\left(\frac{E - E_F}{kT}\right)} = \frac{1}{f(E)} - 1$$

$$\therefore \frac{E - E_F}{kT} = \ln \left(\frac{1}{f(E)} - 1 \right)$$

For $f(E) = 0.99$

$$E = 2.1 + 8.625 \times 10^{-5} \times 300 \ln \left(\frac{1}{0.99} - 1 \right) = 1.981 \text{ eV}$$

For $f(E) = 0.01$

$$E = 2.1 + 8.625 \times 10^{-5} \times 300 \ln \left(\frac{1}{0.01} - 1 \right) = 2.218 \text{ eV}$$

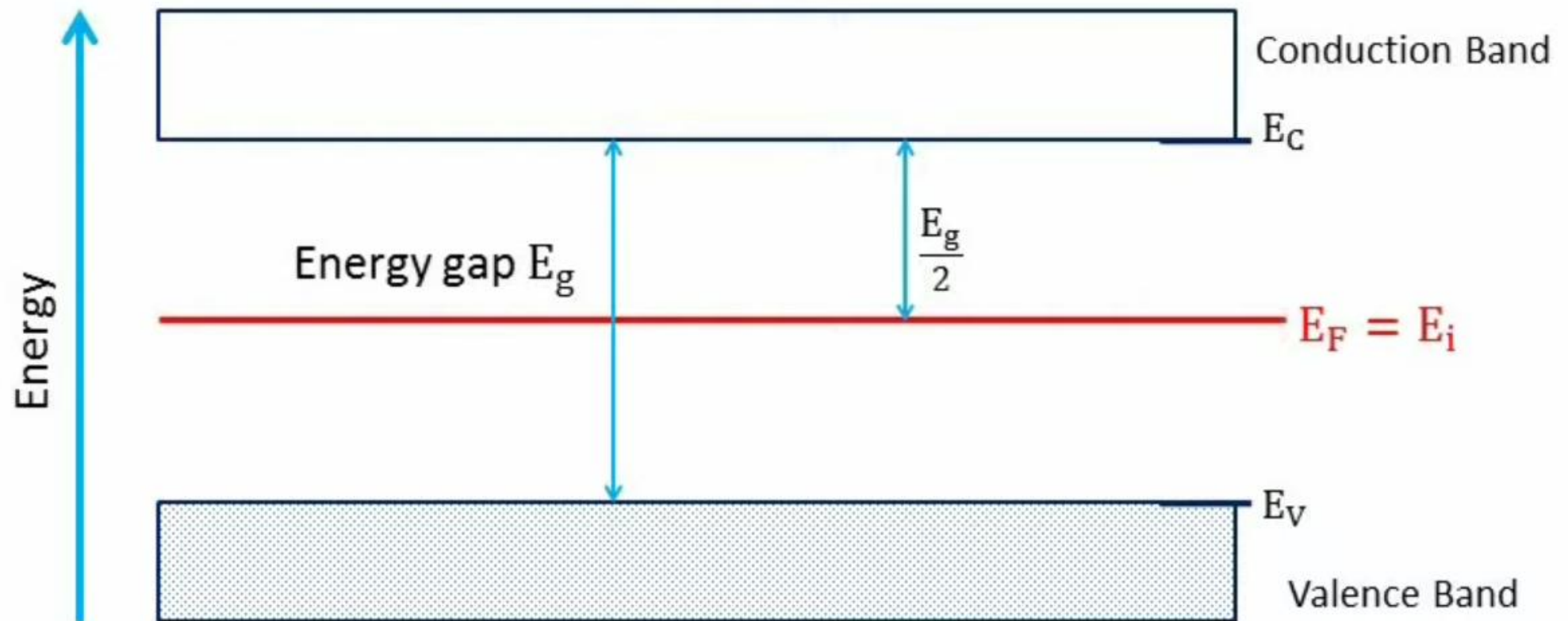
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Fermi level in Semiconductor

- In semiconductors, the **Fermi level** is defined as the reference level that gives the probability of occupancy of states in conduction band as well as unoccupied states in valence band.
- **Fermi level** in semiconductors may also be defined as energy level that corresponds to the centre of gravity of conduction band electrons and valence band holes weighted according to their energies.

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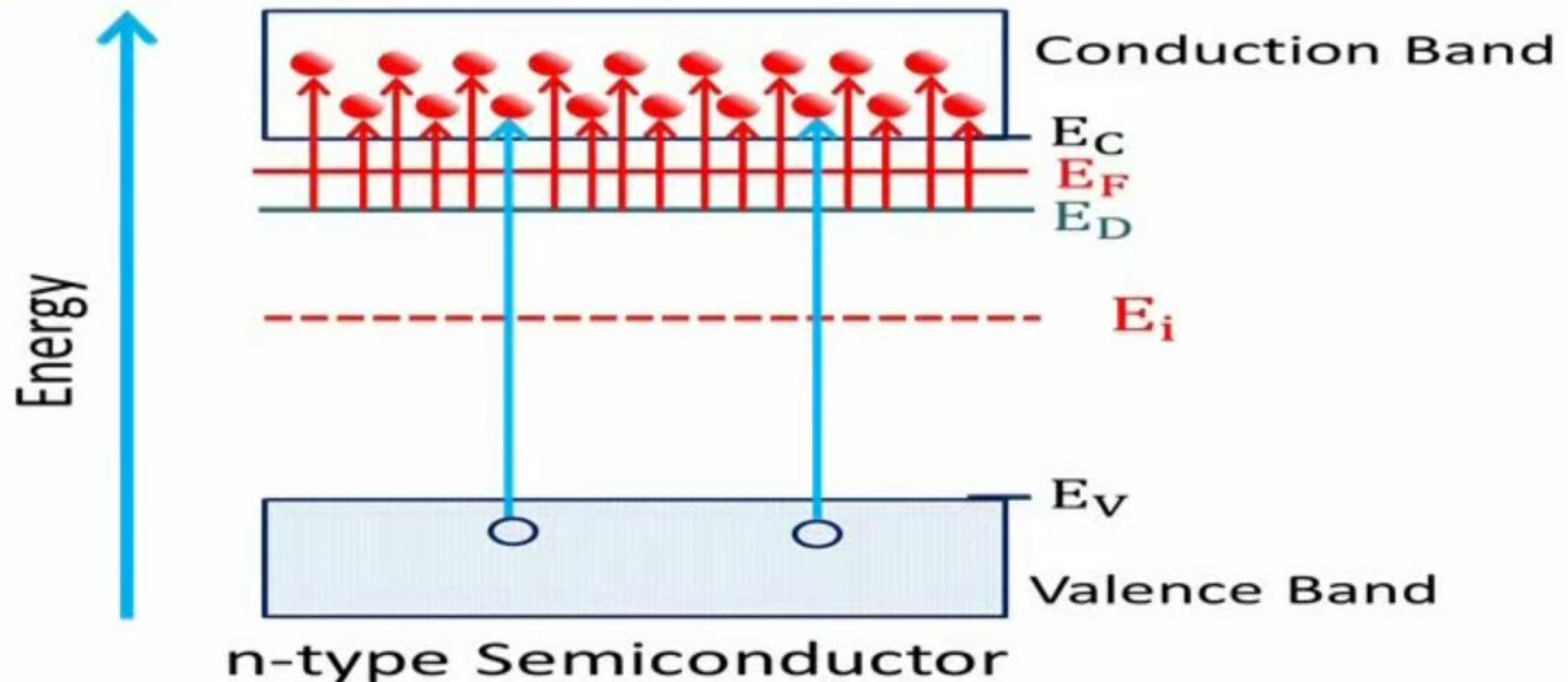
Fermi level in Intrinsic Semiconductor



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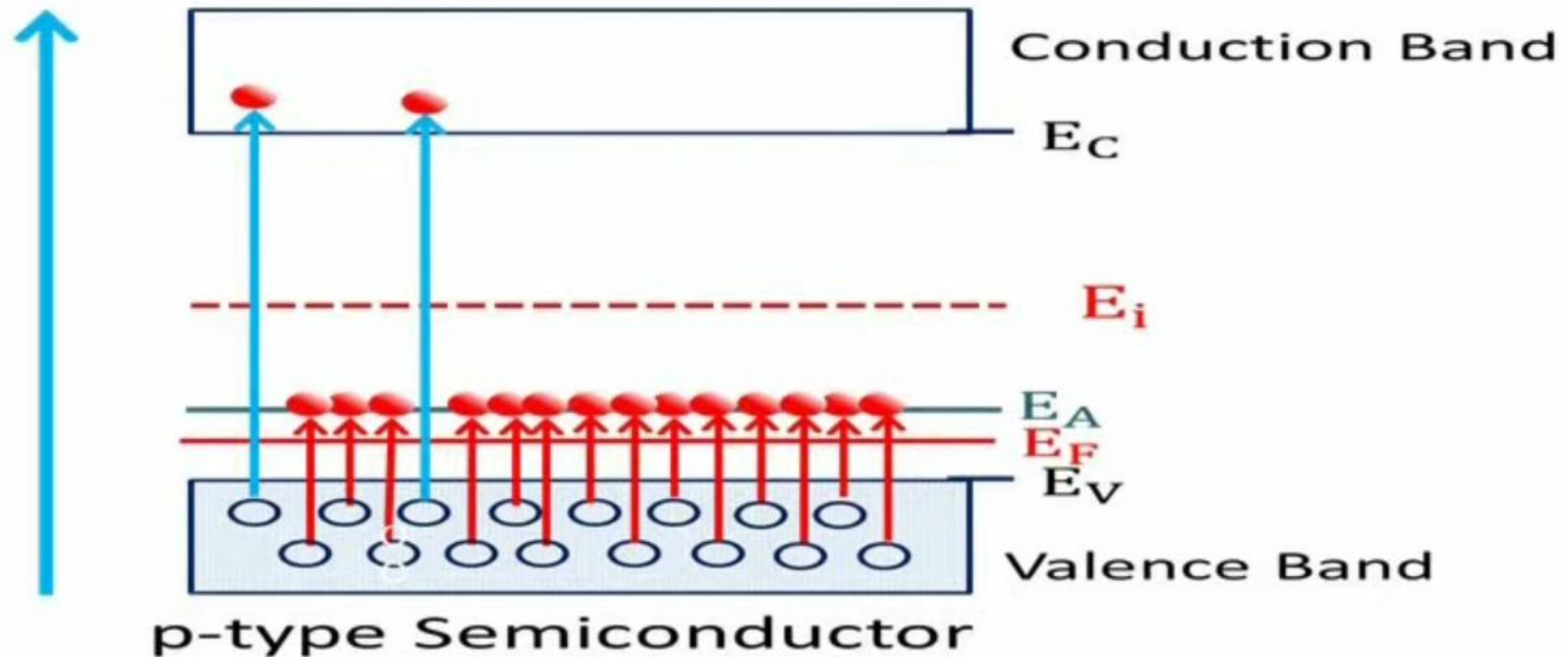
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Fermi level in n-type Semiconductor



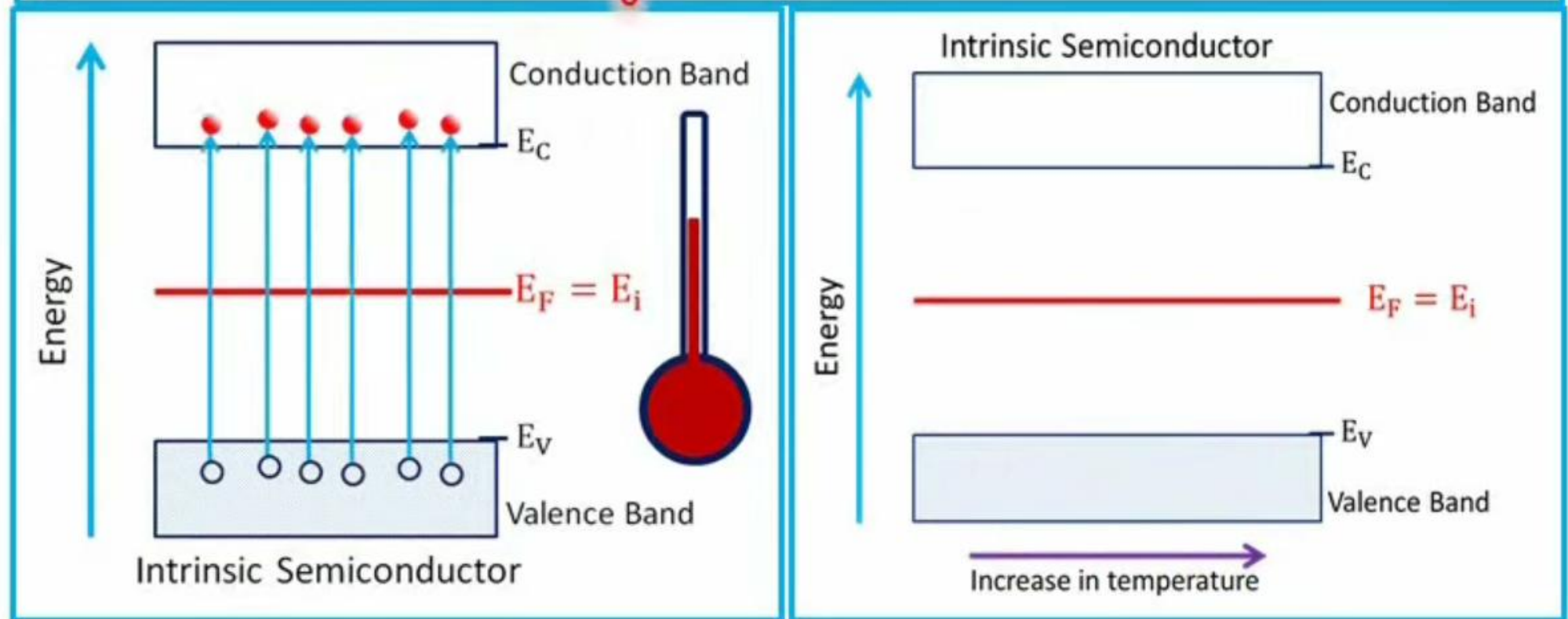
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Fermi level in p-type Semiconductor



Module 3: Semiconductor Physics

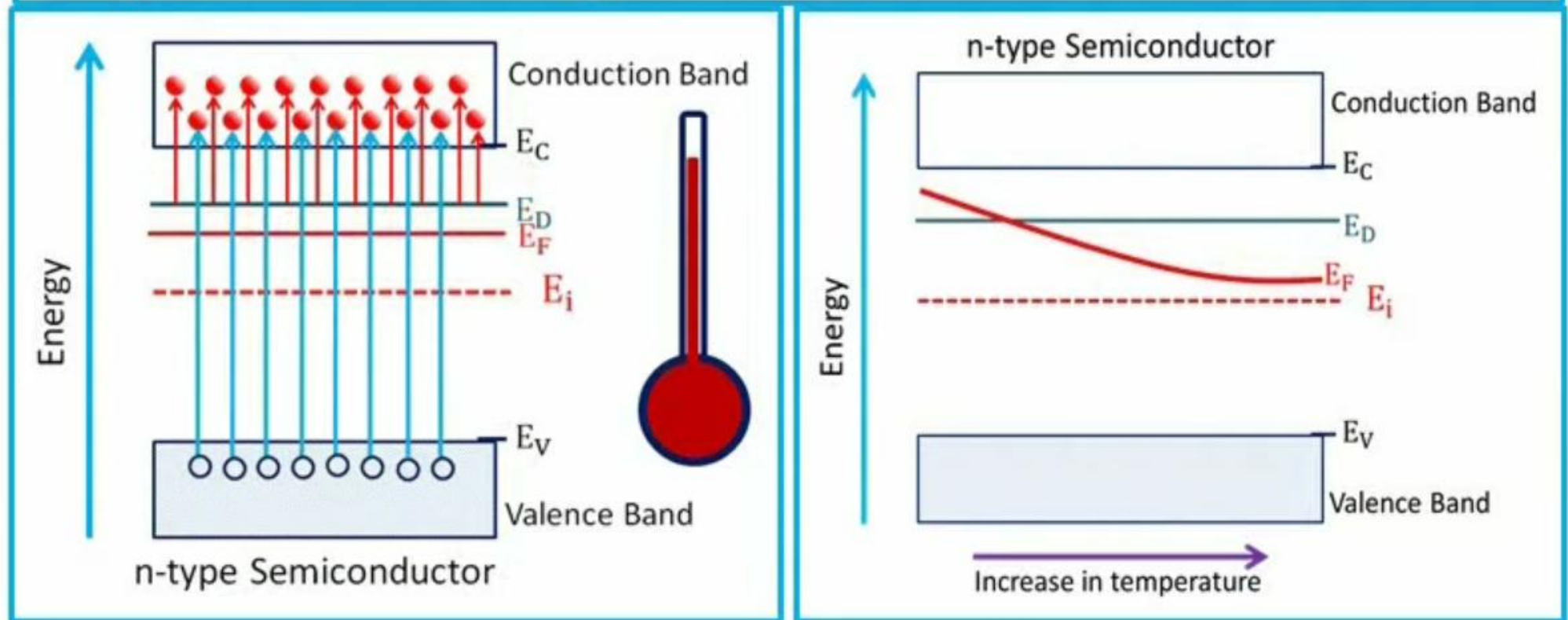
Effect of increasing temperature on Fermi level (Intrinsic Semiconductor)



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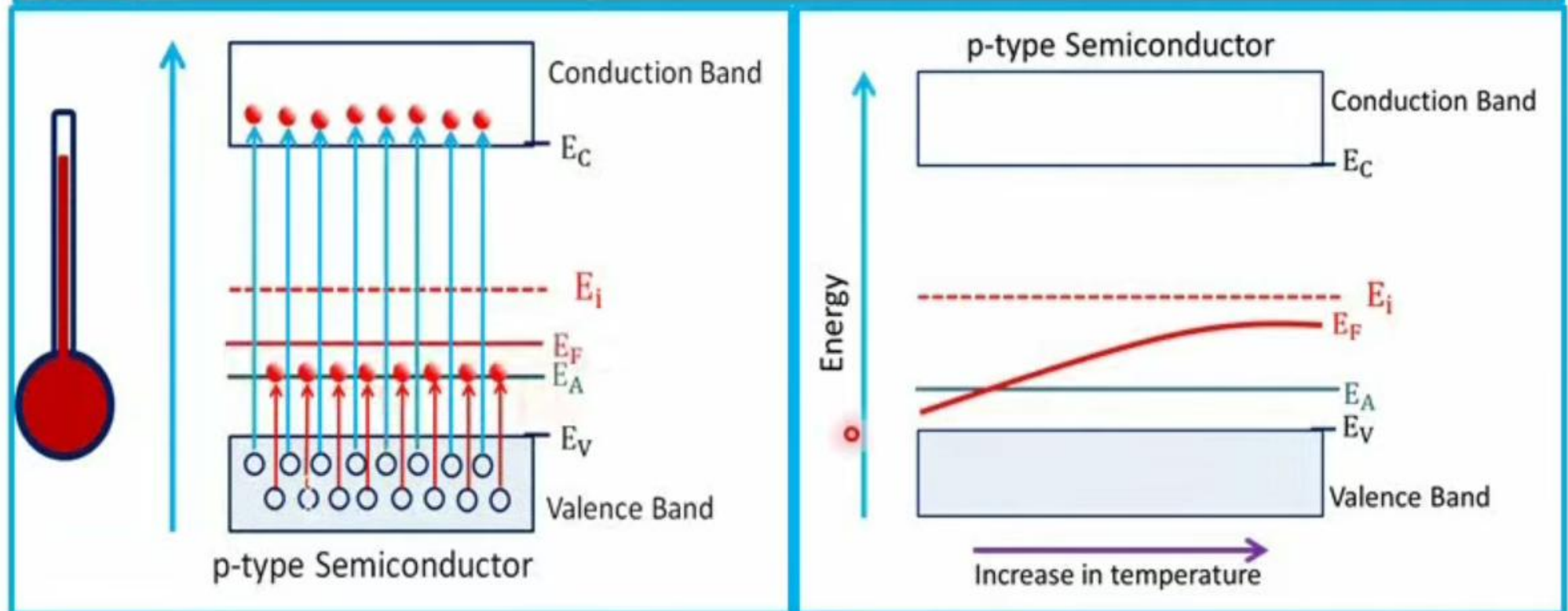
Effect of increasing temperature on Fermi level (n-type Semiconductor)



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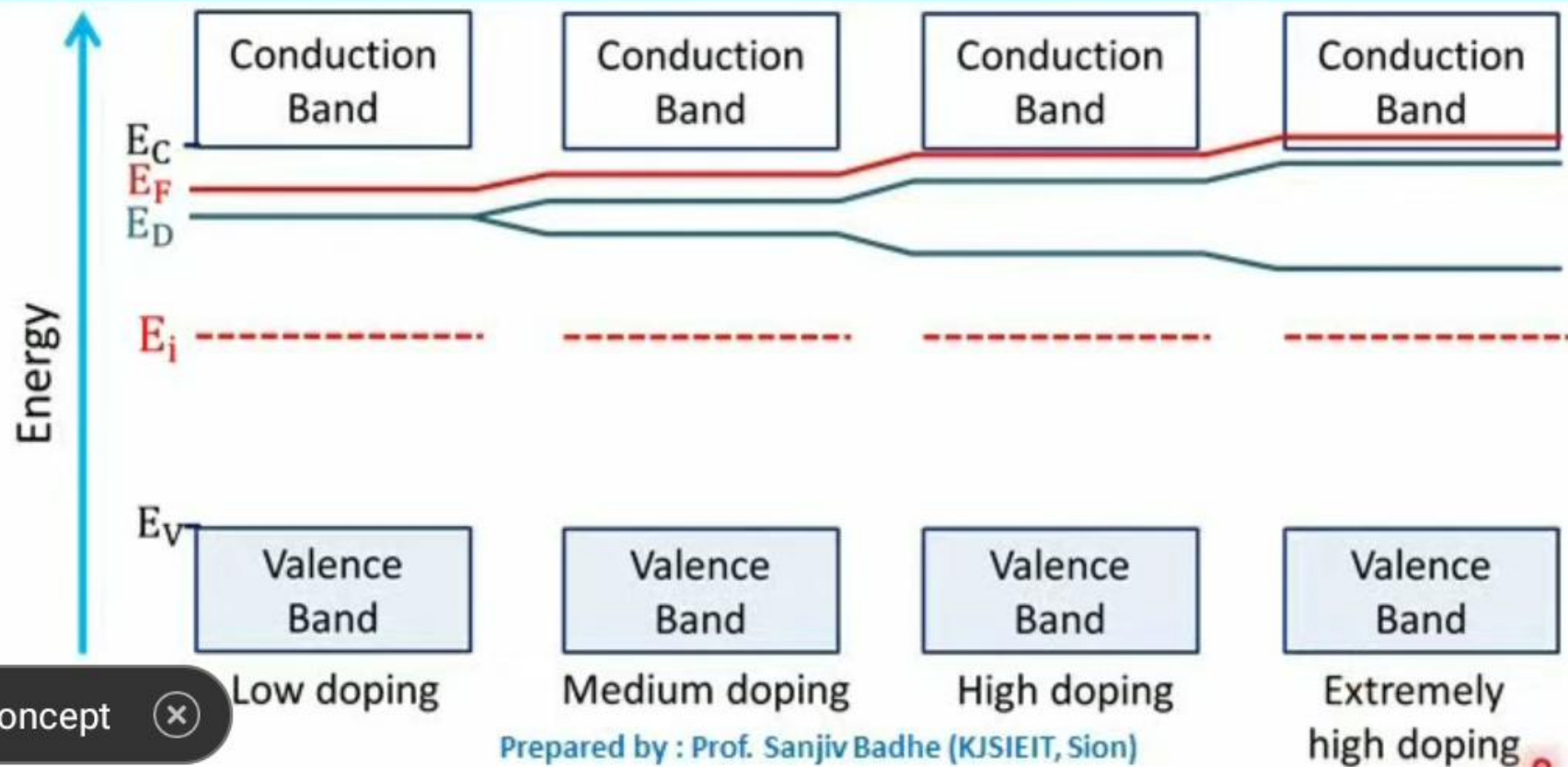
Effect of increasing temperature on Fermi level (p-type Semiconductor)



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Effect of increasing doping concentration on Fermi level
(n-type Semiconductor)



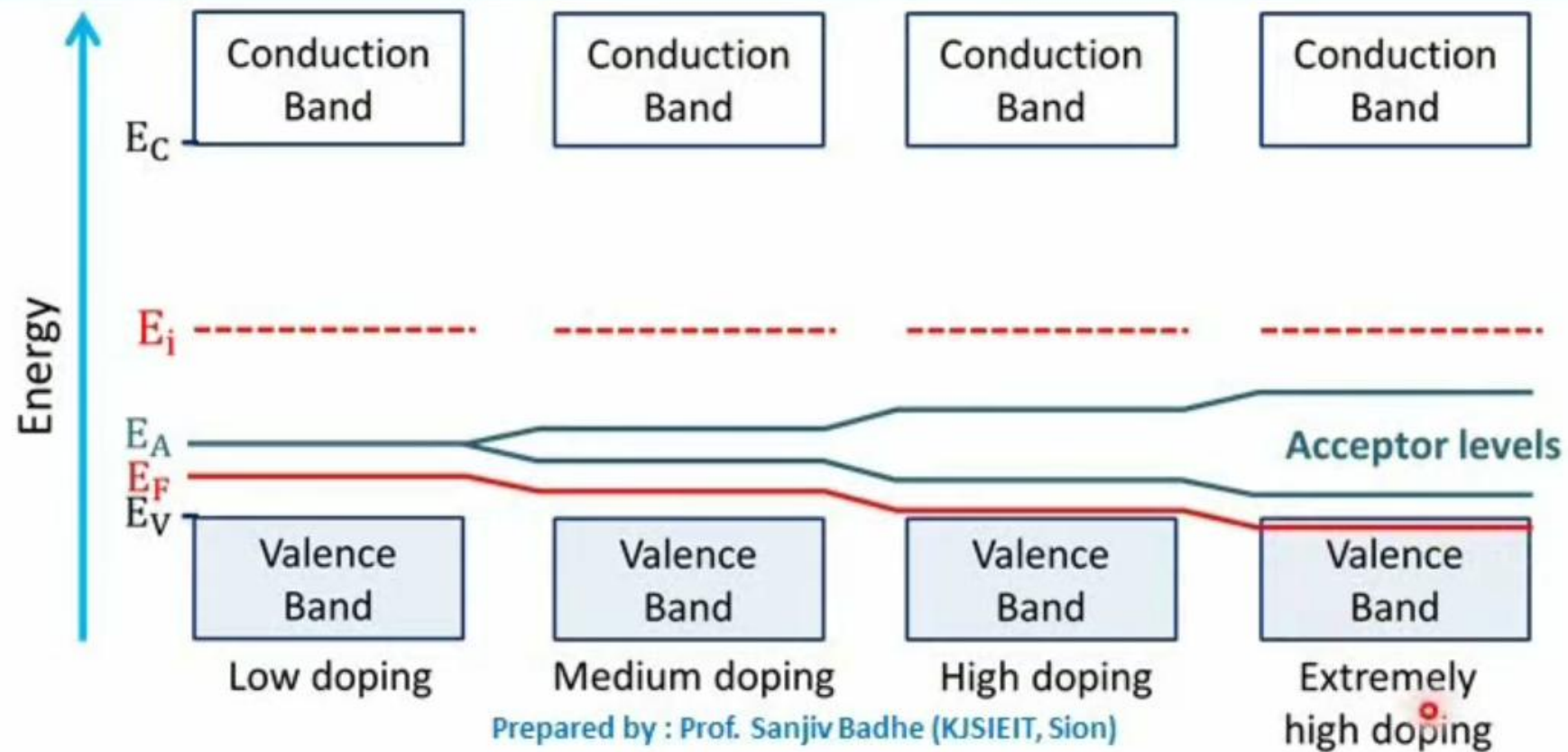
View key concept



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Effect of increasing doping concentration on Fermi level
(p-type Semiconductor)



Module 3: Semiconductor Physics

Carrier Concentration

The Fermi distribution function can be used to calculate the concentration of holes and electrons in the semiconductor, if the density of available states in the valence and conduction bands is known. The concentration of electrons in the conduction band is given by –

$$n = \int_{E_C}^{\infty} f(E)N(E)dE \quad 1$$

The number of electrons per unit volume in the energy range dE is the product of the density of states $N(E)$ and the probability of occupancy $f(E)$. Thus total electron concentration is integral over the entire conduction band.

Module 3: Semiconductor Physics

Carrier Concentration

$$n = \int_{E_C}^{\infty} f(E)N(E)dE \quad (1)$$

Solving equation (1) we get the electron concentration in the conduction band as –

$$n = N_C e^{-(E_C - E_F)/kT} \quad (2)$$

where, $N_C = 2 \left(\frac{2 \pi m_n^* k T}{h^2} \right)^{3/2}$;

m_n^* is the effective mass of the electron

Similarly, hole concentration in the valence band can be obtained as -

$$p = N_V e^{-(E_F - E_V)/kT} \quad (3)$$

where, $N_V = 2 \left(\frac{2 \pi m_p^* k T}{h^2} \right)^{3/2}$;

m_p^* is the effective mass of the hole

At thermal equilibrium, electron and hole concentrations given by equations (2) and (3) are valid for intrinsic as well as extrinsic semiconductor

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Carrier Concentration

$$n = \int_{E_C}^{\infty} f(E)N(E)dE$$

1

$$n = N_C e^{-(E_C - E_F)/kT}$$

2

$$p = N_V e^{-(E_F - E_V)/kT}$$

3

Product of n and p can be written as

$$n \cdot p = N_C e^{-(E_C - E_F)/kT} \cdot N_V e^{-(E_F - E_V)/kT}$$

$$\therefore n \cdot p = N_C N_V e^{-(E_C - E_V)/kT}$$

4

For intrinsic semiconductor,

$$n = n_i \text{ and } p = p_i \text{ and } E_F = E_i$$

$$n_i \cdot p_i = N_C e^{-(E_C - E_i)/kT} \cdot N_V e^{-(E_i - E_V)/kT}$$

$$\therefore n_i \cdot p_i = N_C N_V e^{-(E_C - E_V)/kT}$$

For intrinsic semiconductor, $n_i = p_i$

$$\therefore (n_i)^2 = N_C N_V e^{-(E_C - E_V)/kT}$$

5

Form 4 and 5

$$n \cdot p = (n_i)^2$$

Module 3: Semiconductor Physics

Fermi level in intrinsic semiconductor lies at the middle of the forbidden gap

The **electron concentration in conduction band** of semiconductor at temperature T is given by -

$$n = N_C e^{-(E_C - E_F)/kT} \quad (1)$$

N_C is the effective density of states in conduction band

The **hole concentration in valence band** of semiconductor at temperature T is given by -

$$p = N_V e^{-(E_F - E_V)/kT} \quad (2)$$

N_V is the effective density of states in valence band

For intrinsic semiconductor,

$$n = p$$

$$\therefore N_C e^{-(E_C - E_F)/kT} = N_V e^{-(E_F - E_V)/kT}$$

$$\therefore \frac{e^{-(E_C - E_F)/kT}}{e^{-(E_F - E_V)/kT}} = \frac{N_C}{N_V}$$

$$\text{At equilibrium, } N_C = N_V \quad (3)$$

$$\therefore \frac{e^{-(E_C - E_F)/kT}}{e^{-(E_F - E_V)/kT}} = 1$$

Module 3: Semiconductor Physics

Fermi level in intrinsic semiconductor lies at the middle of the forbidden gap

$$n = N_C e^{-(E_C - E_F)/kT} \quad 1$$

$$p = N_V e^{-(E_F - E_V)/kT} \quad 2$$

For intrinsic semiconductor,

$$n = p$$

$$\therefore N_C e^{-(E_C - E_F)/kT} = N_V e^{-(E_F - E_V)/kT}$$

$$\text{At equilibrium, } N_C = N_V \quad 3$$

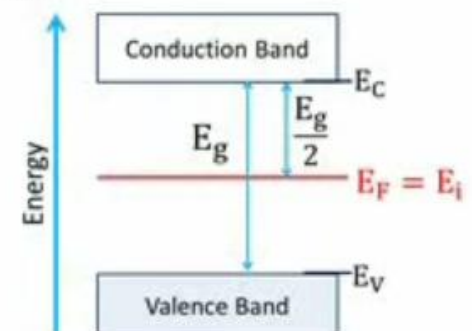
$$\therefore \frac{e^{-(E_C - E_F)/kT}}{e^{-(E_F - E_V)/kT}} = 1$$

$$\therefore e^{\frac{-(E_C - E_F - E_F + E_V)}{kT}} = 1 = e^0$$

$$\therefore \frac{E_C - E_F - E_F + E_V}{kT} = 0$$

$$\therefore E_C + E_V = 2E_F$$

$$\therefore E_F = \frac{E_C + E_V}{2}$$



What is the probability of an electron being thermally promoted to conduction band in diamond at 27°C if band gap is 5.6 eV wide.

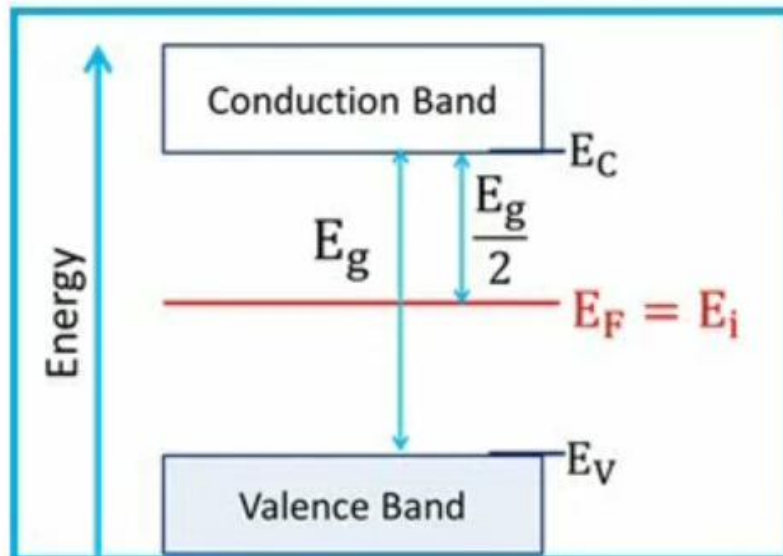
Given : $T = 27^{\circ}\text{C} = 300^{\circ}\text{K}$,

$$k = 8.625 \times 10^{-5} \frac{\text{eV}}{^{\circ}\text{K}},$$

$$E_g = 5.6 \text{ eV}$$

For diamond, $E_F = \frac{E_C + E_V}{2}$

$$\therefore E_C - E_F = \frac{E_g}{2} = \frac{5.6}{2} = 2.8 \text{ eV}$$



$$\therefore \text{Probability of occupancy} = f(E_C) = \frac{1}{1 + e^{\left(\frac{E_C - E_F}{kT}\right)}}$$

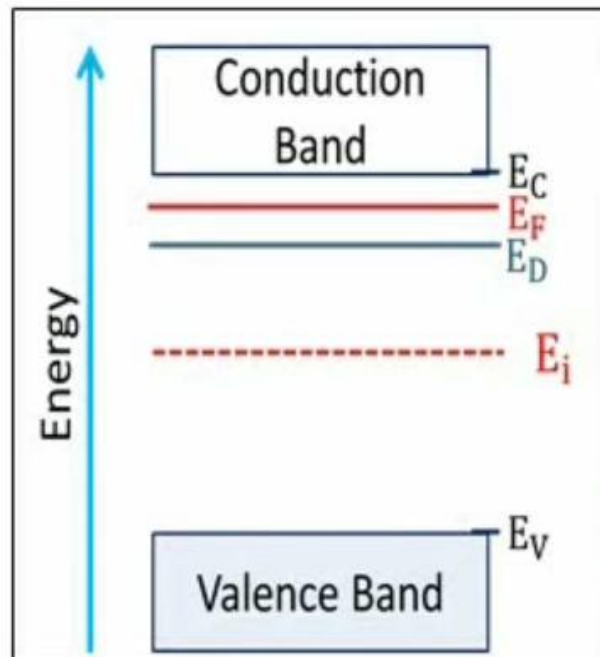
$$= \frac{1}{1 + e^{\left(\frac{2.8}{8.625 \times 10^{-5} \times 300}\right)}}$$

$$= 1.09 \times 10^{-47}$$

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In an n-type semiconductor, the Fermi level lies 0.4 eV below the conduction band. If the concentration of donor atoms is doubled, find the new position of the Fermi level w.r.t. conduction band.

Given : $E_C - E_F = 0.4 \text{ eV}$



Temperature is not mentioned.
Let us assume $T = 300 \text{ K}$

Electron concentration is given by

$$n = N_C e^{-(E_C - E_F)/kT} \quad (1)$$

When donor concentration is doubled, electron concentration will also get doubled and fermi level will be shifted to E_{F2}

$$2n = N_C e^{-(E_C - E_{F2})/kT} \quad (2)$$

Divide (2) by (1)

$$\frac{e^{-(E_C - E_{F2})/kT}}{e^{-(E_C - E_F)/kT}} = 2$$

$$\therefore e^{\frac{-(E_C - E_{F2}) + (E_C - E_F)}{kT}} = 2$$

$$\therefore -(E_C - E_{F2}) + (E_C - E_F) = kT \ln(2)$$

$$\therefore (E_C - E_{F2}) = (E_C - E_F) - kT \ln(2)$$

$$\therefore (E_C - E_{F2}) = 0.4 - 8.625 \times 10^{-5} \times 300 \ln(2)$$

$$\therefore (E_C - E_{F2}) = 0.382 \text{ eV}$$

So, The Fermi level shifts towards the Conduction band as donor concentration increases.

Module 3: Semiconductor Physics

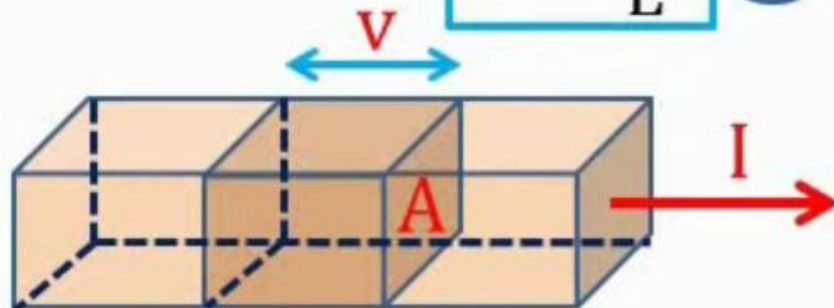
Conductivity and Mobility

Drift Current Density is given by -

$$J = \sigma E \quad (1)$$

Mobility is given by -

$$\mu = \frac{v}{E} \quad (2)$$



Let v is drift velocity of electrons

$\therefore$ length traversed by the electrons
in unit time = v

$\therefore$ Volume swept by electrons in unit time = $v A$

If ' n ' is number of electrons per unit volume,
number of electrons in volume $vA = n v A$

If e is charge on electron, charge flowing per
unit time is -

$$\text{current } I = n e v A$$

$\therefore$ Current Density,

$$J = \frac{I}{A} = \frac{n e v A}{A} = n e v$$

From (1) and (2)

$$\text{Conductivity, } \sigma = \frac{J}{E} = \frac{n e v}{E} = n e \mu$$

Module 3: Semiconductor Physics

Conductivity of Semiconductors

Conductivity of semiconductor is given by -

$$\sigma = \sigma_e + \sigma_h$$

$$\therefore \sigma = n e \mu_e + p e \mu_h$$

Where,

n is electron concentration in conduction band

p is hole concentration in valence band

e is charge on electron

μ_e is mobility of electrons

μ_h is mobility of holes

View key concept



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Conductivity of **Intrinsic** Semiconductors

Conductivity of semiconductor is given by -

$$\therefore \sigma = n e \mu_e + p e \mu_h$$

For intrinsic semiconductor, **$n = p = n_i$**

$\therefore$ Conductivity of **intrinsic semiconductors** is given by -

$$\therefore \sigma_i = n_i e \mu_e + n_i e \mu_h$$

$$\therefore \sigma_i = n_i e (\mu_e + \mu_h)$$

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Conductivity of Extrinsic Semiconductors

Conductivity of semiconductor is given by -

$$\therefore \sigma = n e \mu_e + p e \mu_h$$

For n-type semiconductor, $n \gg p$

$\therefore$ Conductivity of **n-type semiconductors** is given by -

$$\sigma = n e \mu_e = N_d e \mu_e$$

N_d is donor concentration

For p-type semiconductor, $p \gg n$

$\therefore$ Conductivity of **p-type semiconductors** is given by -

$$\sigma = p e \mu_h = N_a e \mu_h$$

N_a is acceptor concentration

Find resistivity of Copper assuming that each atom contributes one free electron for conduction. Given density of Cu = 8.96 gm / cm³, atomic weight = 63.5,

Avogadro's Number = 6.023×10^{23} / gm-mol, Mobility of electron = 43.3 cm²/V-sec.

Given : $\mu_e = 43.3$ cm²/V-sec, density $\rho = 8.96$ gm/cm³, Atomic weight = 63.5

Resistivity = ?

In 63.5 gm there are 6.023×10^{23} atoms

Number of atoms per unit volume (i.e. in 8.96 gm) will be –

$$\text{Atomic density} = \frac{8.96 \times 6.023 \times 10^{23}}{63.5} = 8.4985 \times 10^{22} / \text{cm}^3$$

each atom contributes one free electron

$$\therefore \text{electron concentration} = n = 1 \times \text{Atomic density} = 8.4985 \times 10^{22} / \text{cm}^3$$

Resistivity is given by –

$$\rho = \frac{1}{\sigma} = \frac{1}{n e \mu_e} = \frac{1}{8.4985 \times 10^{22} \times 1.6 \times 10^{-19} \times 43.3} = 1.698 \times 10^{-6} \Omega - \text{cm}$$

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Calculate conductivity of a germanium sample if donor impurity atoms are added to the extent of one part in 10^8 germanium atoms at room temperature. Assume that only one electron of each atom takes part in conduction process.

Avogadro's number = 6.023×10^{23} / gm-mol, Density of Ge = 5.32 gm / cm^3

Atomic weight of Ge = 72.6, mobility of electrons = $3800 \text{ cm}^2/\text{volt-sec}$

72.6 gm Ge contains 6.023×10^{23} atoms

$\therefore$ 5.32 gm i.e. 1 cm^3 will have $\frac{6.023 \times 10^{23}}{72.6} \times 5.32 = 4.41 \times 10^{22}$ atoms

As donor impurity is 1 part in 10^8 , donor concentration, $N_d = \frac{4.41 \times 10^{22}}{10^8}$
 $= 4.41 \times 10^{14} / \text{cm}^3$

As only one electron of each donor atom takes part in conduction,
electron concentration = $n = 4.41 \times 10^{14} / \text{cm}^3$

Now, conductivity, $\sigma = n e \mu_e = 4.41 \times 10^{14} \times 1.6 \times 10^{-19} \times 3800 = 0.268 / \Omega \text{ cm}$

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Silicon sample is doped with 10^{17} As atoms/cm³. Intrinsic concentration for Silicon at room temperature is 1.5×10^{10} /cm³. What is the equilibrium hole concentration at room temperature? Show the position of Fermi level.

Given : $n = N_d = 10^{17}$ /cm³

$n_i = 1.5 \times 10^{10}$ /cm³

hole concentration = $p = ?$

position of Fermi level ?

For a semiconductor,

$$n \cdot p = (n_i)^2$$

Therefore, hole concentration is given by

$$p = \frac{(n_i)^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{17}} = 2.25 \times 10^3 \text{ /cm}^3$$

We know, $n = N_C e^{-(E_C - E_F)/kT}$ 1

Let the middle of the forbidden gap be E_i .

This is the Fermi level in intrinsic semiconductor

∴ For intrinsic semiconductor, $n_i = N_C e^{-(E_C - E_i)/kT}$ 2

From 1 and 2

$$n = N_C e^{-(E_C - E_i)/kT} e^{(E_F - E_i)/kT}$$

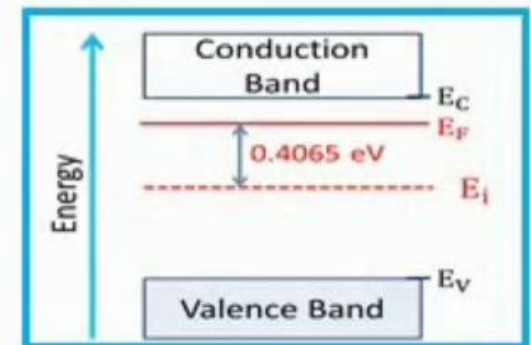
$$= n_i e^{(E_F - E_i)/kT}$$

$$\therefore \frac{n}{n_i} = e^{(E_F - E_i)/kT}$$

$$\therefore E_F - E_i = kT \ln \left(\frac{n}{n_i} \right)$$

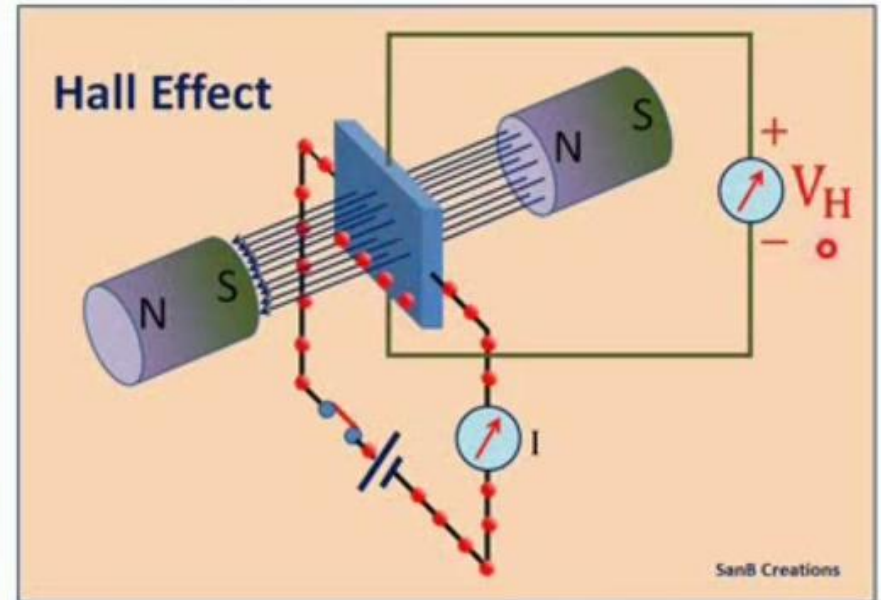
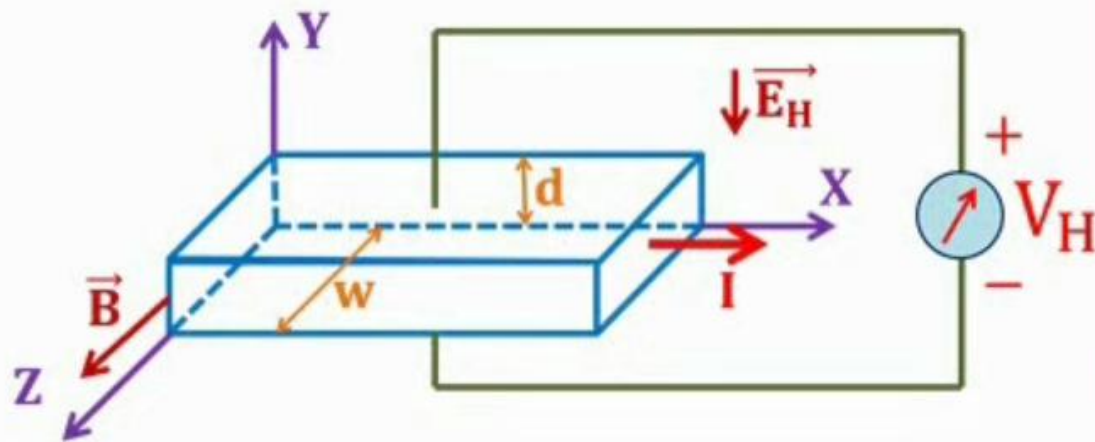
$$= 8.625 \times 10^{-5} \times 300 \times \ln \left(\frac{10^{17}}{1.5 \times 10^{10}} \right)$$

$$= 0.4065 \text{ eV}$$



Module 3: Semiconductor Physics

Hall Effect

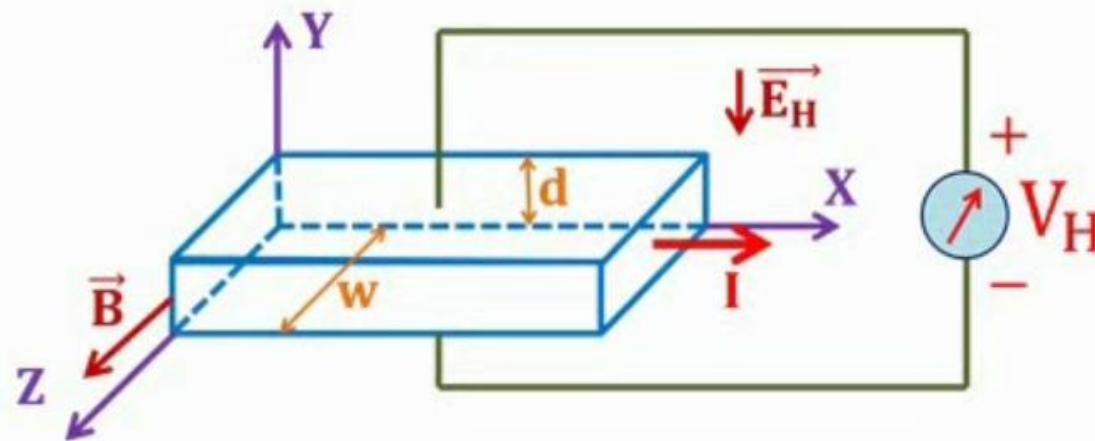


If a metal or semiconductor carrying a current I is placed in a transverse magnetic field B , an electric field E is induced in the direction perpendicular to both the direction of current and magnetic field. This phenomenon is called as **Hall Effect**.

The electric field produced is called as **Hall field (E_H)**.

Module 3: Semiconductor Physics

Hall Effect



The current through the metal or semiconductor is given by –

$$I = n e v A \quad (1)$$

where n is concentration of electrons,
 A is area of cross section of the plate,
 e is charge on the electron,
 v is average drift velocity of electrons.

At equilibrium,
electric force on electrons = magnetic force on electrons

$$e E_H = e v B \quad (2)$$

$E_H = \frac{V_H}{d}$ is the Hall field and
 V_H is Hall voltage

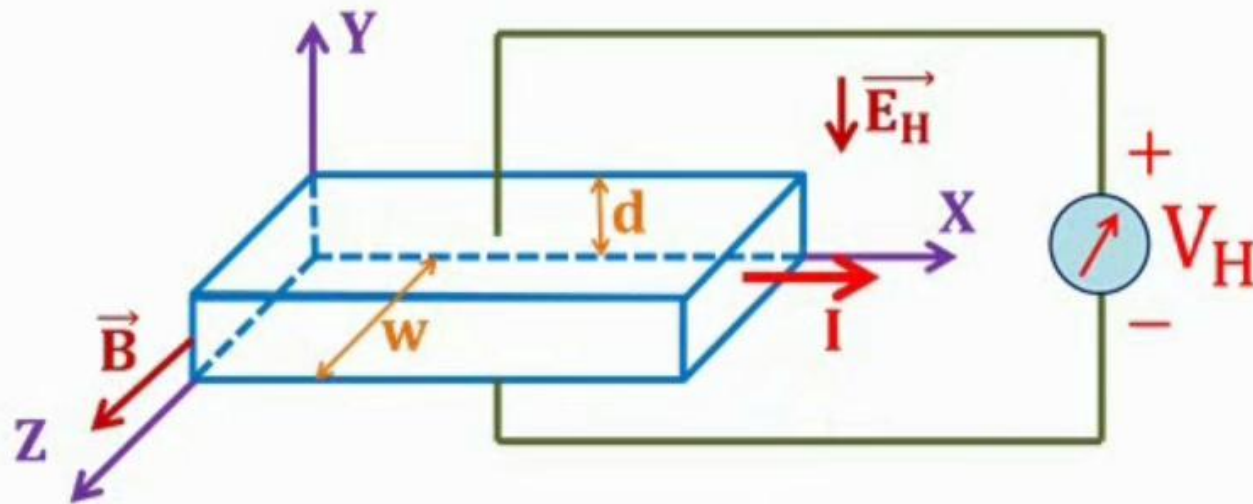
View key concept



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Module 3: Semiconductor Physics

Hall Effect



$$I = n e v A \quad (1)$$

$$e E_H = e v B \quad (2)$$

$$\therefore \frac{V_H}{d} = v B$$

$$\text{As } E_H = \frac{V_H}{d}$$

$$\therefore V_H = v B d$$

$$\therefore V_H = \frac{n e v B d}{n e}$$

From

1

$$\therefore V_H = \frac{I B d}{n e A} = \frac{I B d w}{n e A w}$$

$$\therefore V_H = \frac{B I}{n e w} \quad (3)$$

Module 3: Semiconductor Physics

Hall Effect

$$I = n e v A \quad (1)$$

$$e E_H = e v B \quad (2)$$

$$\therefore V_H = \frac{B I}{n e w} \quad (3)$$

$$\therefore V_H = R_H \frac{B I}{w}$$

$R_H = \frac{1}{ne}$ is Hall Coefficient

$$\therefore R_H = \frac{1}{ne} = \frac{V_H w}{B I} = \frac{V_H d w}{d B I} = \frac{E_H A}{B I} = \frac{E_H}{B J}$$

$\therefore$ Hall coefficient is defined as the Hall field per unit magnetic induction per unit current density.

Importance / Significance of Hall Effect is that it helps –

- To determine carrier concentration (n).
- To determine type of the semiconductor.
- To determine mobility μ of the charge carriers if conductivity of the semiconductor is known. $\sigma = ne\mu$

In a Hall effect experiment, a potential difference of $4.5 \mu\text{V}$ is developed across a foil of zinc of thickness 0.02 mm , while carrying a current of 1.5 A in a direction perpendicular to applied magnetic field of 2 tesla . Calculate a) Hall coefficient for zinc
b) concentration of electrons.

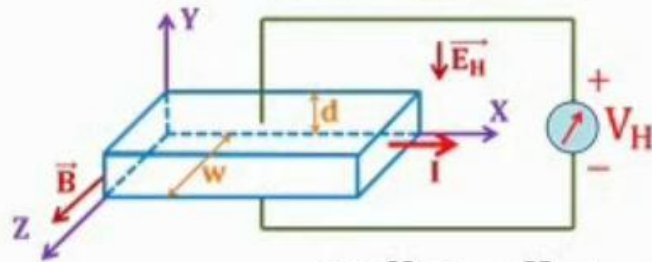
Given : $V_H = 4.5 \mu\text{V} = 4.5 \times 10^{-6} \text{ V}$

$I = 1.5 \text{ A}$

$B = 2 \text{ T}$

thickness = $w = 0.02 \text{ mm} = 0.02 \times 10^{-3} \text{ m}$

$R_H = ? \quad n = ?$



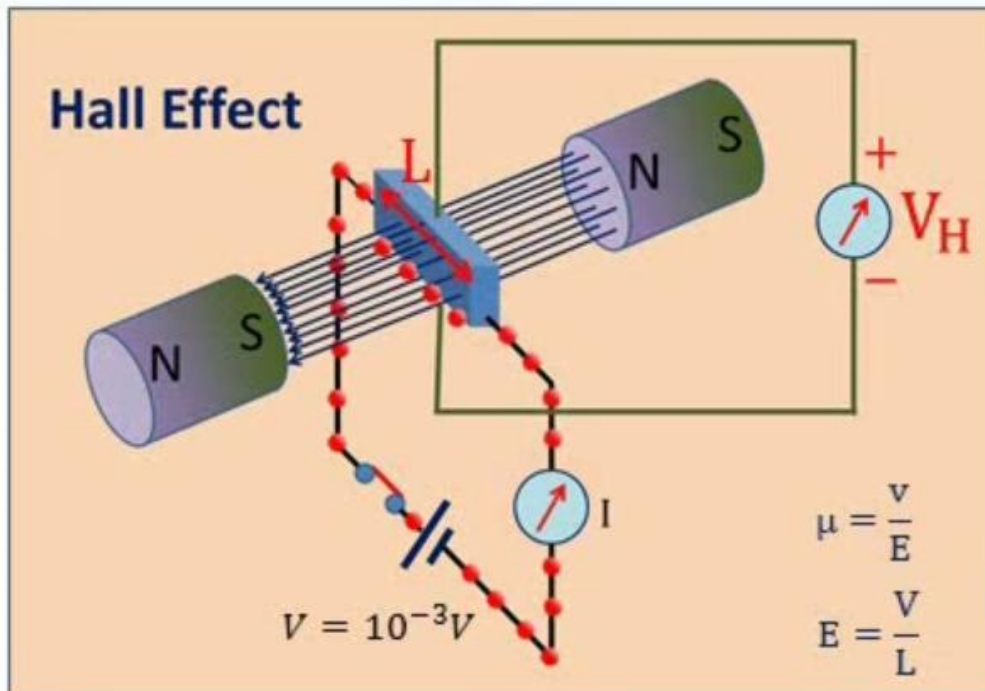
As $V_H = \frac{B I}{n e w}$

Hall Coefficient = $R_H = \frac{1}{n e} = \frac{V_H w}{B I} = \frac{4.5 \times 10^{-6} \times 0.02 \times 10^{-3}}{2 \times 1.5} = 3 \times 10^{-11} \frac{\text{m}^3}{\text{C}}$

Concentration of electrons = $n = \frac{B I}{V_H e w}$

$= \frac{2 \times 1.5}{4.5 \times 10^{-6} \times 1.6 \times 10^{-19} \times 0.02 \times 10^{-3}} = 2.083 \times 10^{29} / \text{m}^3$

A bar of n-type Ge of size $0.010\text{m} \times 0.001\text{m} \times 0.001\text{m}$ is mounted in a magnetic field of $2 \times 10^{-1}\text{T}$. The electron density in the bar is $7 \times 10^{21} / \text{m}^3$. If one millivolt is applied across the long ends of the bar, determine the current through the bar and the voltage between Hall electrodes placed across the short dimensions of the bar. Assume mobility of electrons = $0.39 \frac{\text{m}^2}{\text{V-sec}}$



Given : $n = 7 \times 10^{21} / \text{m}^3$ $d = w = 0.001 \text{ m}$
 $B = 0.2 \text{ T}$ $V = 10^{-3} \text{ V}$
 $L = 0.010 \text{ m}$ $I = ?$ $V_H = ?$

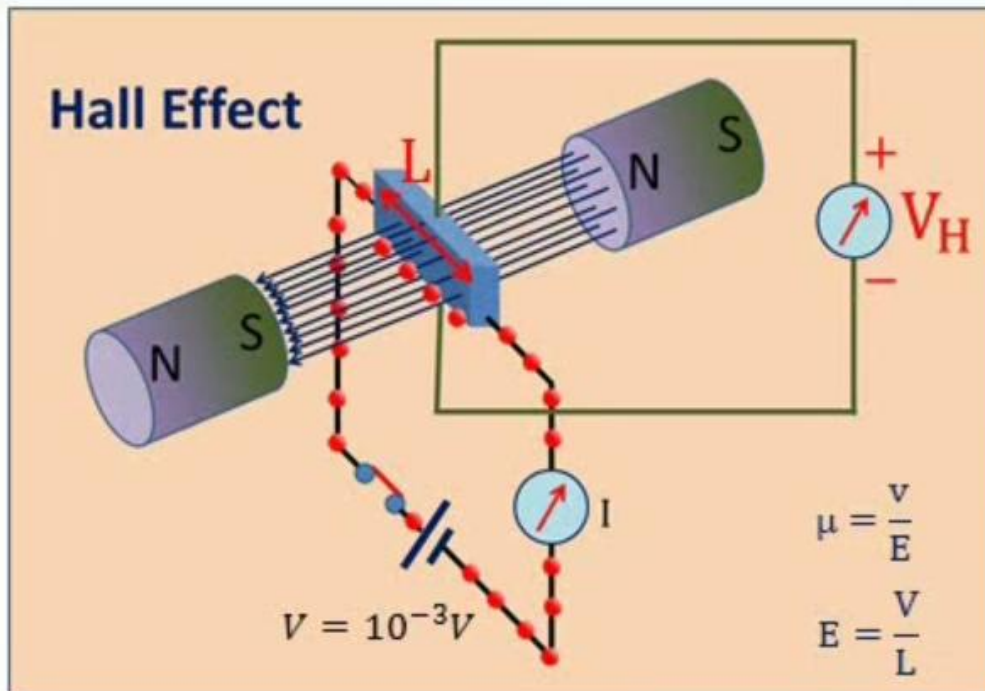
Current through the bar = $I = n e v A$

$$\therefore I = n e \mu E d w = n e \mu \frac{V}{L} d w$$

$$= 7 \times 10^{21} \times 1.6 \times 10^{-19} \times 0.39 \times \frac{10^{-3}}{0.010} \times 0.001 \times 0.001$$

$$\therefore I = 43.68 \times 10^{-6} \text{ A}$$

A bar of n-type Ge of size $0.010\text{m} \times 0.001\text{m} \times 0.001\text{m}$ is mounted in a magnetic field of $2 \times 10^{-1}\text{T}$. The electron density in the bar is $7 \times 10^{21} / \text{m}^3$. If one millivolt is applied across the long ends of the bar, determine the current through the bar and the voltage between Hall electrodes placed across the short dimensions of the bar. Assume mobility of electrons = $0.39 \frac{\text{m}^2}{\text{V}\cdot\text{sec}}$



Given : $n = 7 \times 10^{21} / \text{m}^3$
 $B = 0.2 \text{ T}$
 $L = 0.010 \text{ m}$

$d = w = 0.001 \text{ m}$
 $V = 10^{-3} \text{ V}$
 $I = 43.68 \times 10^{-6} \text{ A}$

Hall Voltage = $V_H = \frac{B I}{n e w}$

$\therefore V_H = \frac{0.2 \times 43.68 \times 10^{-6}}{7 \times 10^{21} \times 1.6 \times 10^{-19} \times 0.001}$
 $= 7.8 \times 10^{-6} \text{ V}$

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Diffusion current



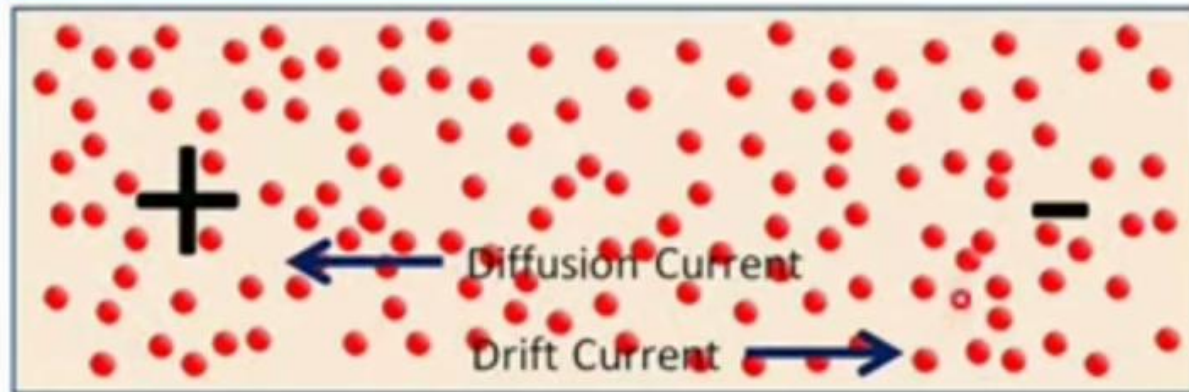
At equilibrium,

Diffusion current = Drift Current

- If excess carriers are introduced locally within a conductor, either by heating or injecting carriers, a non-uniform distribution is created.
- The excess carriers move from region of higher concentration to the region of lower concentration in order to produce uniform distribution.
- **This phenomenon is called as diffusion.**

Module 3: Semiconductor Physics

Diffusion current



At equilibrium,

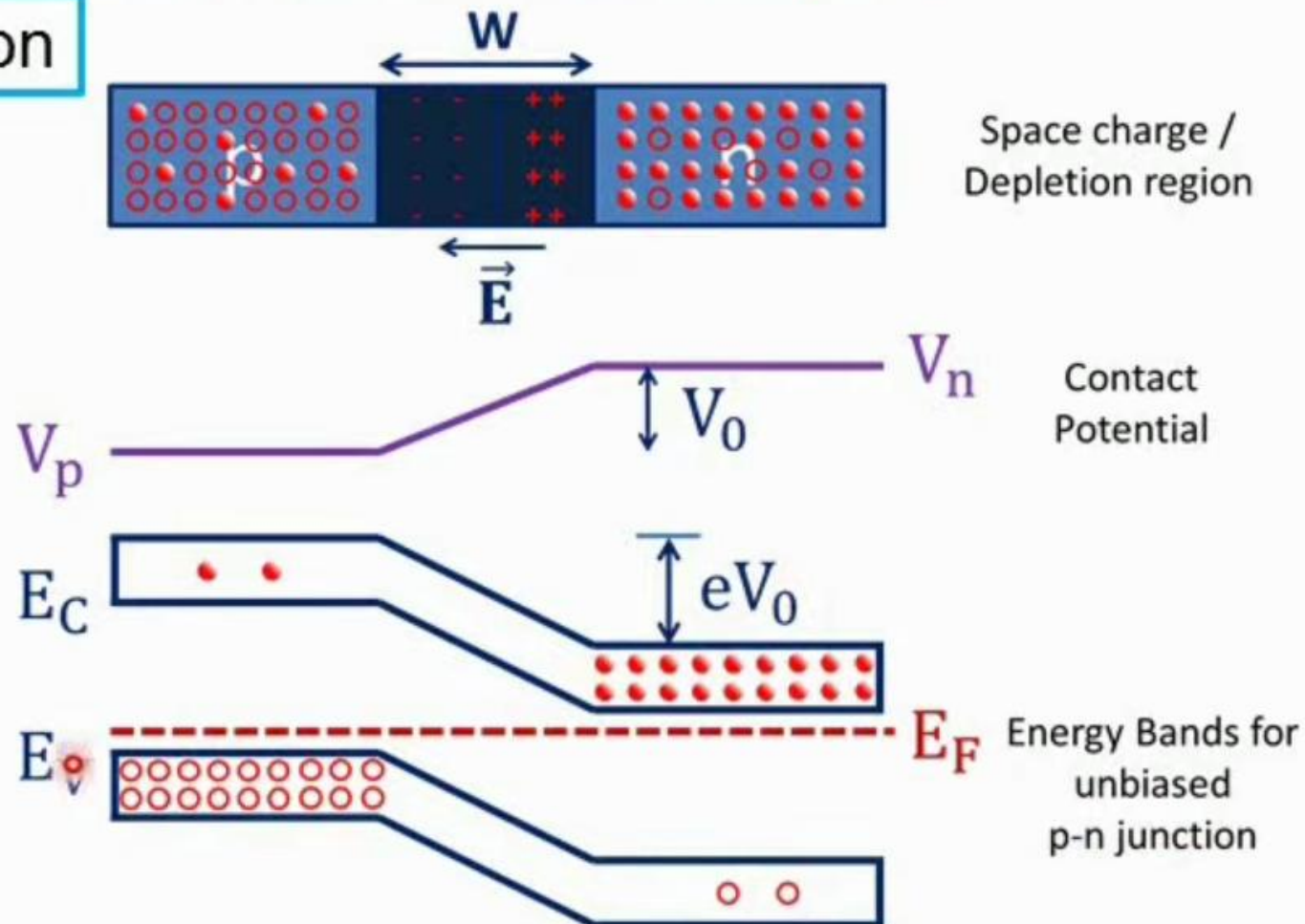
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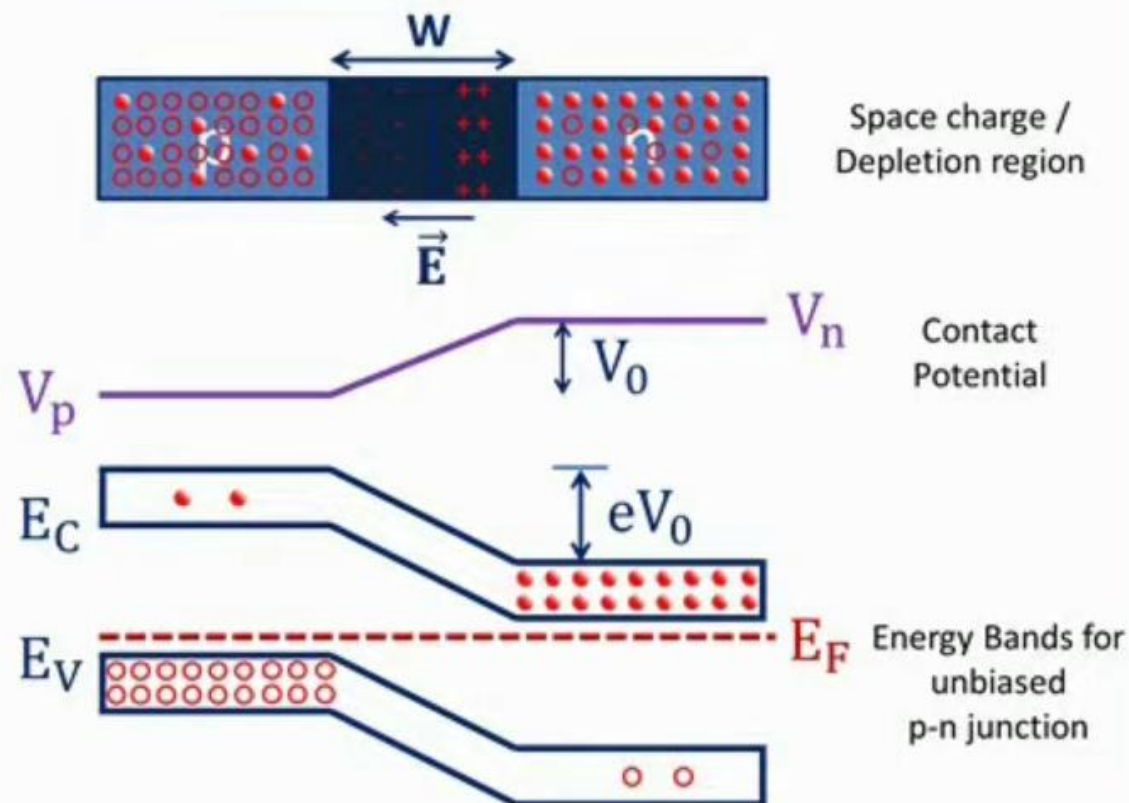
p-n Junction



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p-n Junction



At equilibrium,

$$J_p (\text{drift}) + J_p (\text{diffusion}) = 0$$

$$J_n (\text{drift}) + J_n (\text{diffusion}) = 0$$

Contact potential is given by the relation

$$V_0 = \frac{kT}{e} \ln \left(\frac{N_a N_d}{(n_i)^2} \right)$$

Where

k = Boltzmann constant = $8.625 \times 10^{-5} \text{ eV} / ^\circ\text{K}$,

T = Temperature in $^\circ\text{K}$,

e = charge of electron = $1.6 \times 10^{-19} \text{ C}$,

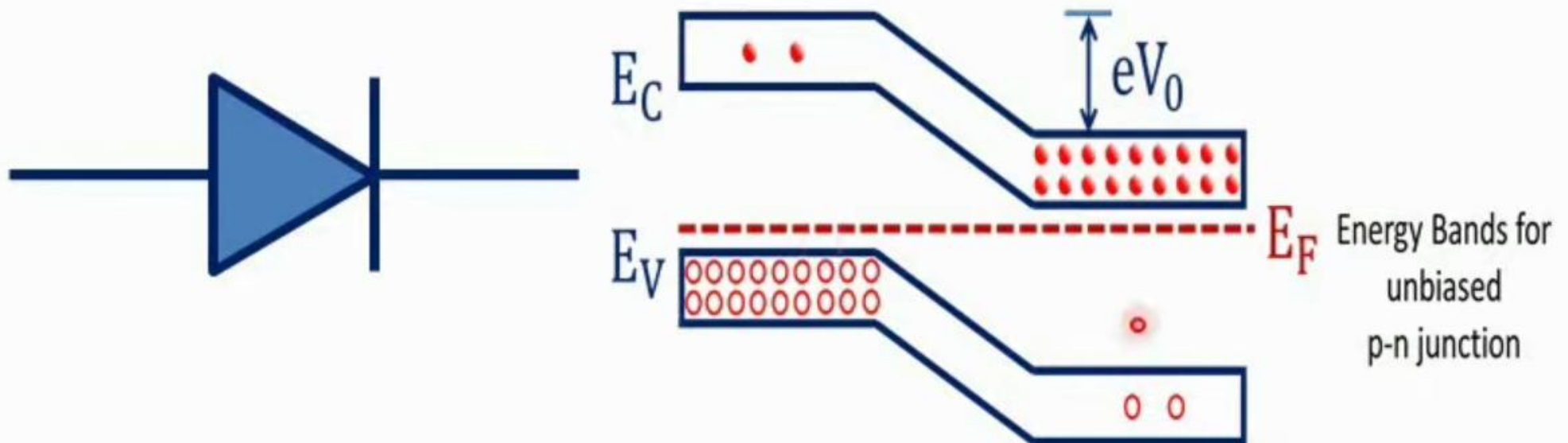
N_a = acceptor concentration on p-side,

N_d = donor concentration on n-side

Module 3: Semiconductor Physics

p-n Junction Diode

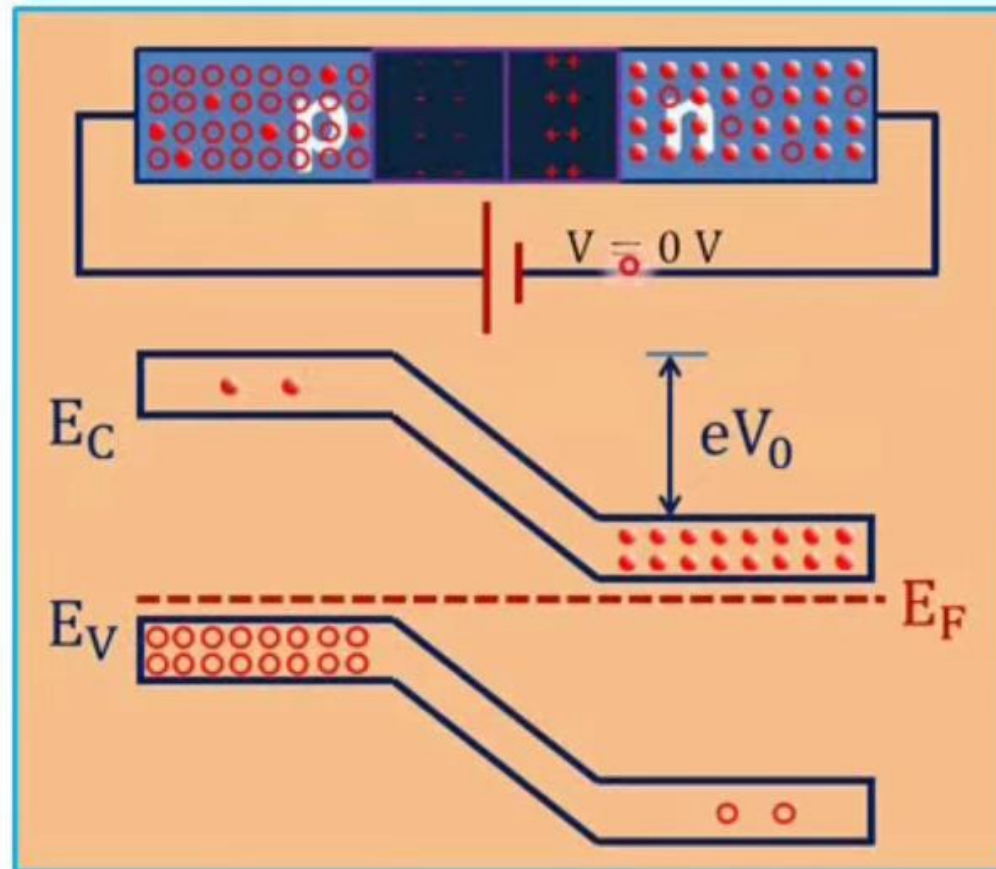
A **diode** is an electrical device allowing current to move through it in one direction with far greater ease than in the other.



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p-n Junction Diode (Forward Bias)

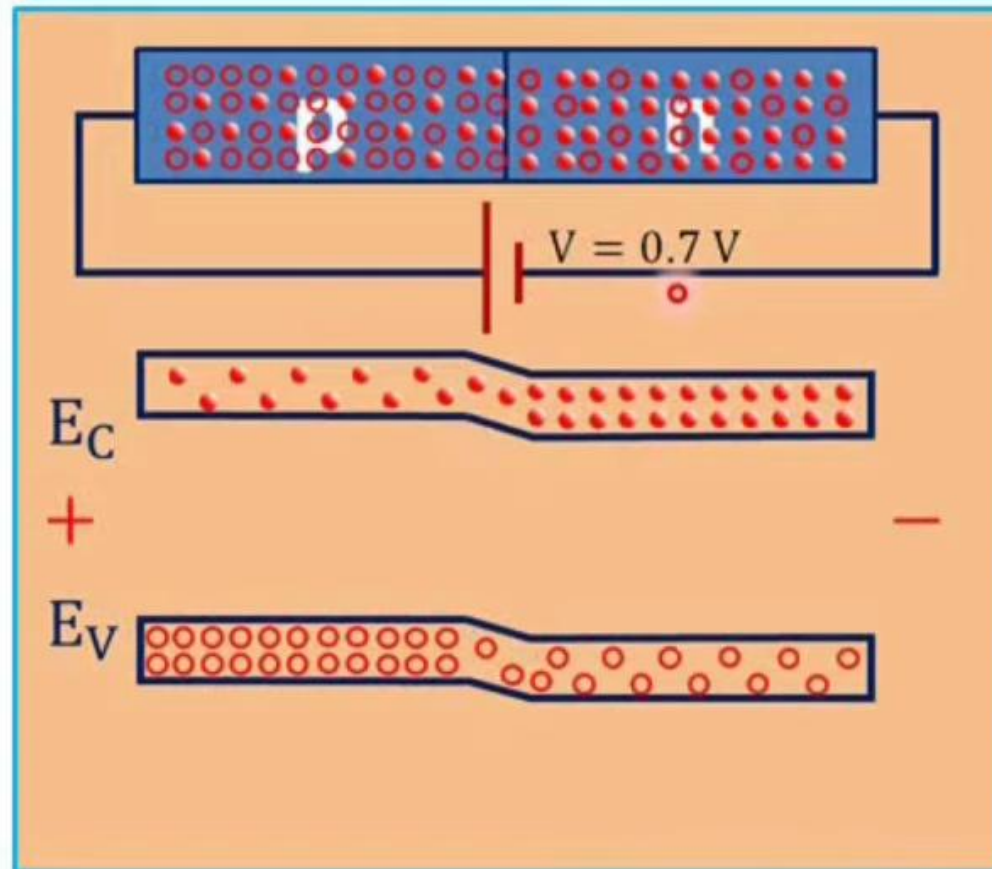


- As forward bias **voltage is increased**, **depletion layer width reduces**.
- When forward bias voltage is equal to barrier voltage, **the depletion layer vanishes** and majority carriers can cross the junction.

- As forward bias voltage is increased, **energy bands on n-side are raised up**.
- When forward bias voltage is equal to barrier voltage, **the energy bands on two sides are aligned** and majority carriers can cross the junction.

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p-n Junction Diode (Forward Bias)

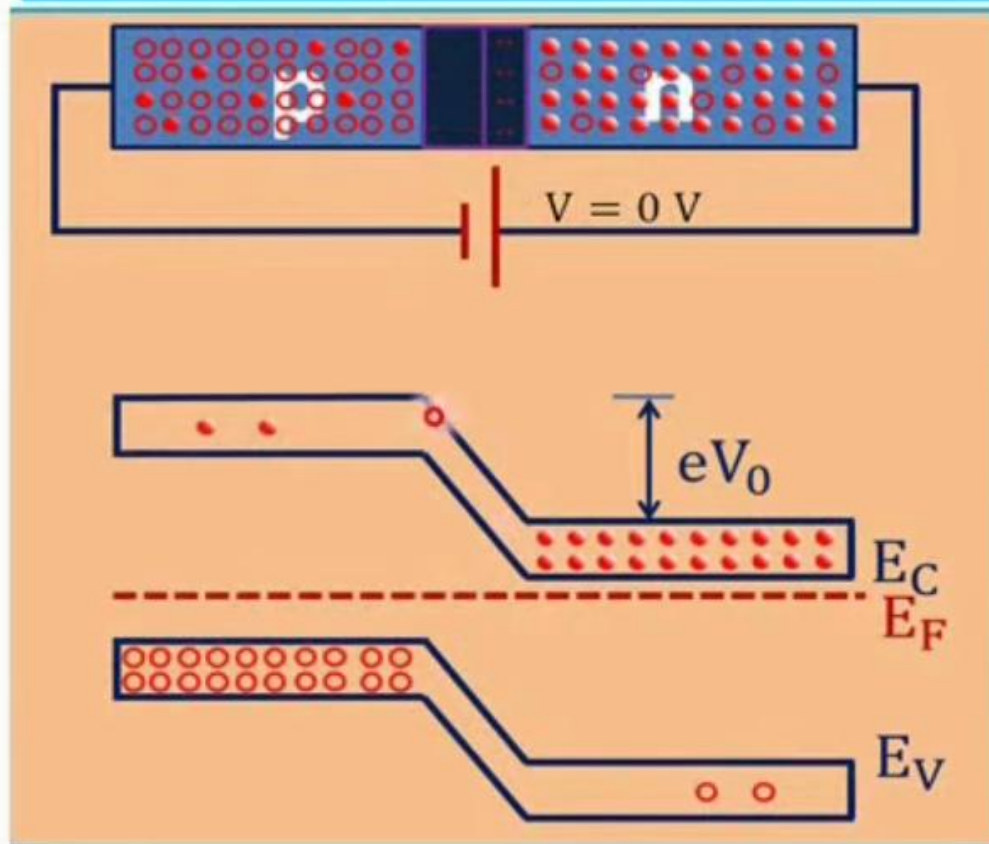


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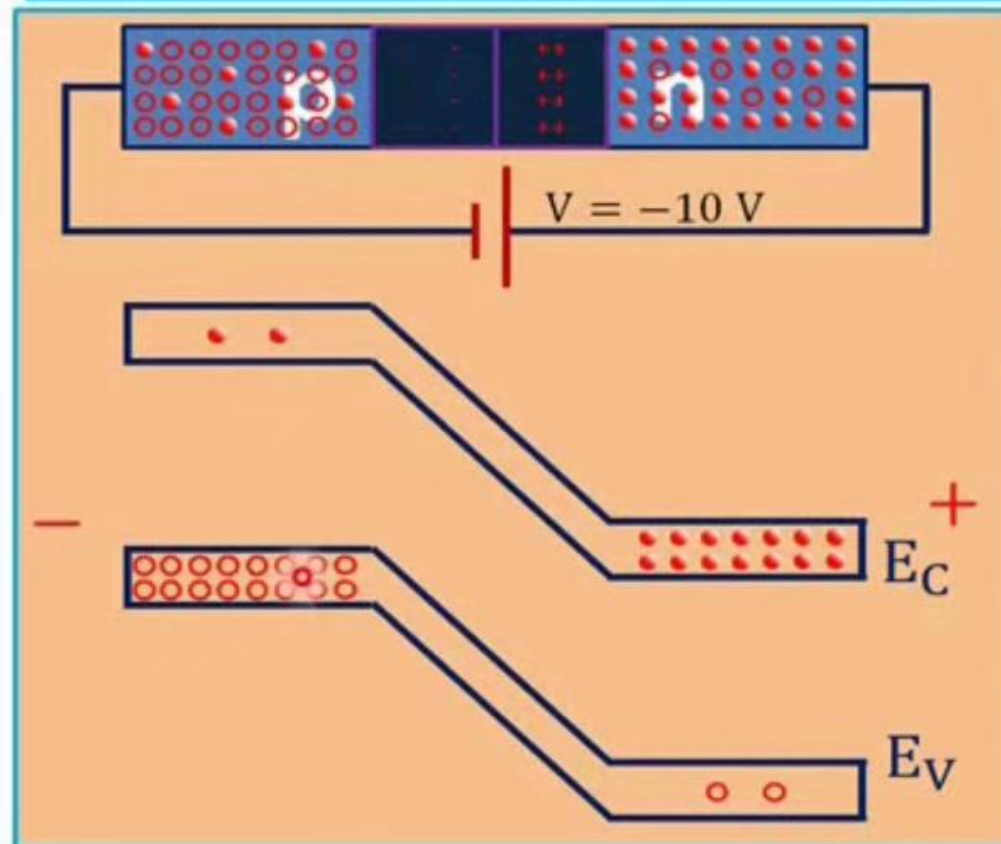
p-n Junction Diode (reverse bias)



- As reverse bias **voltage is increased**, **depletion layer width increases**.
- Majority carriers can not cross the junction as repulsive force increases.
- As reverse bias voltage is increased, **energy bands on p-side are raised up**.
- Height of the energy hill increases and becomes more difficult for the majority carriers on both sides to cross the junction.
- Current due to minority carriers can be obtained which is negligible.

Module 3: Semiconductor Physics

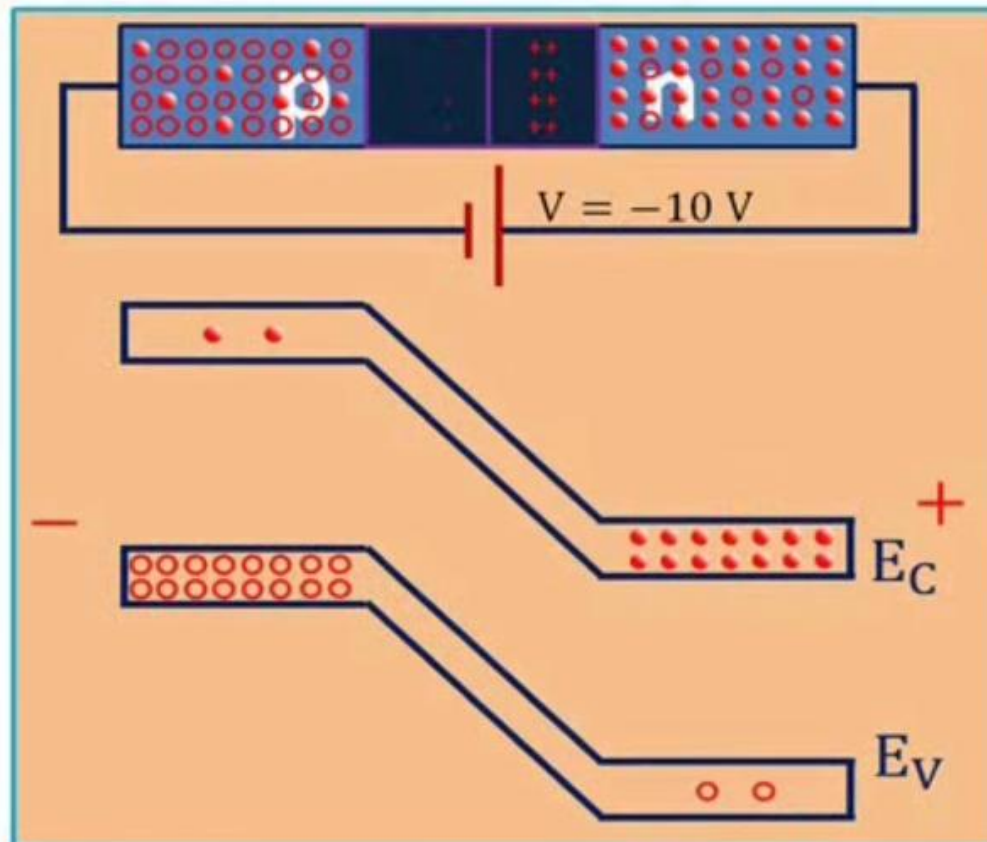
p-n Junction Diode (reverse bias)



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-
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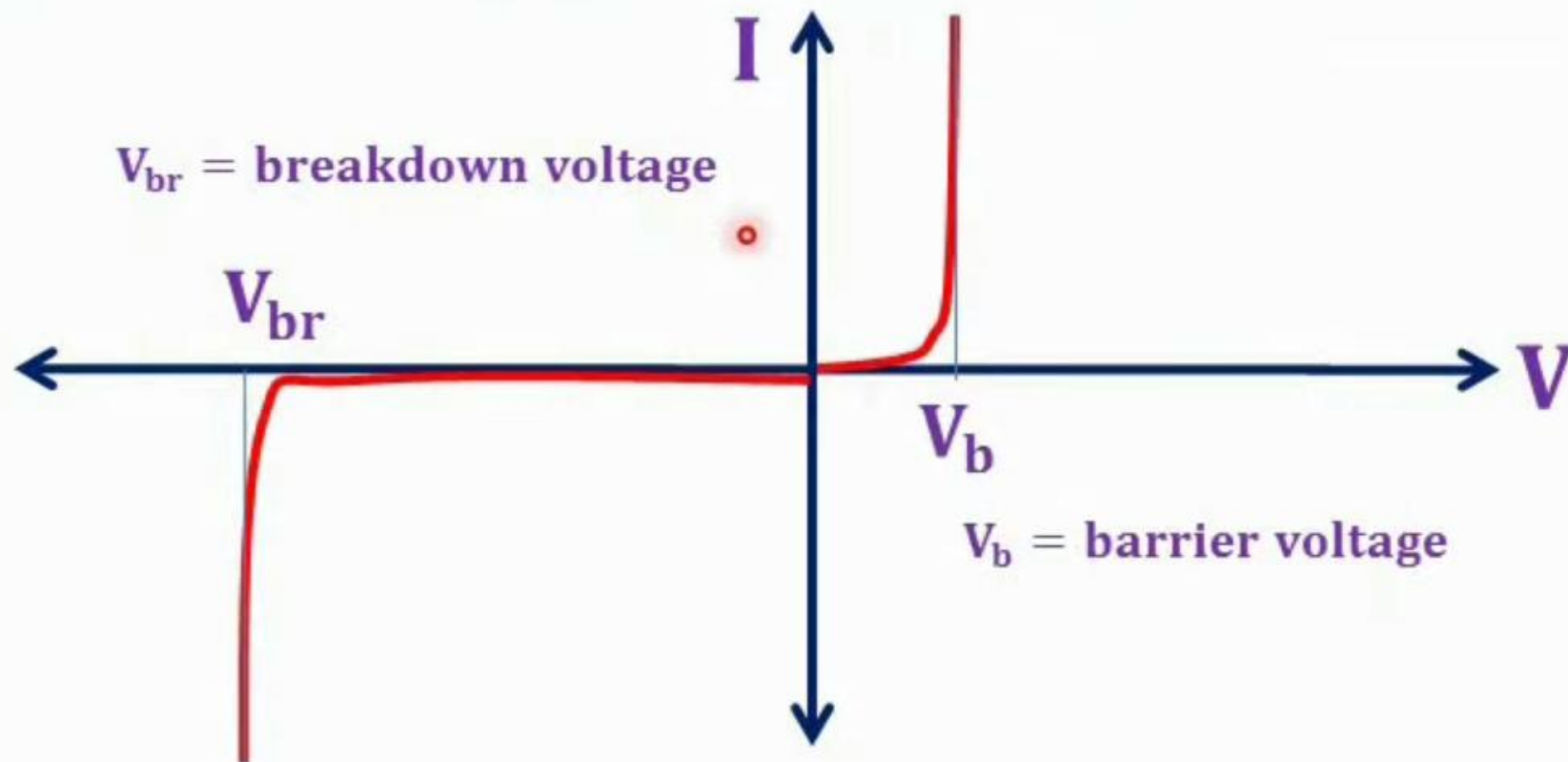
p-n Junction Diode (Breakdown)



- In reverse-bias, only a small current (due to minority carrier) flows through the diode.
- If the reverse bias voltage is very high, then the depletion region widens, and the electric field is very strong.
- The minority charge carriers get accelerated and gain kinetic energy.
- They knock off, other electrons from the negative ions in the depletion layer.
- These new electrons can knock off more electron from the negative ions
- The number of electrons in the depletion layer suddenly gets multiplied.
- This leads to sudden rise in current
- **This is Avalanche Effect**

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p-n Junction Diode (V-I Characteristics)



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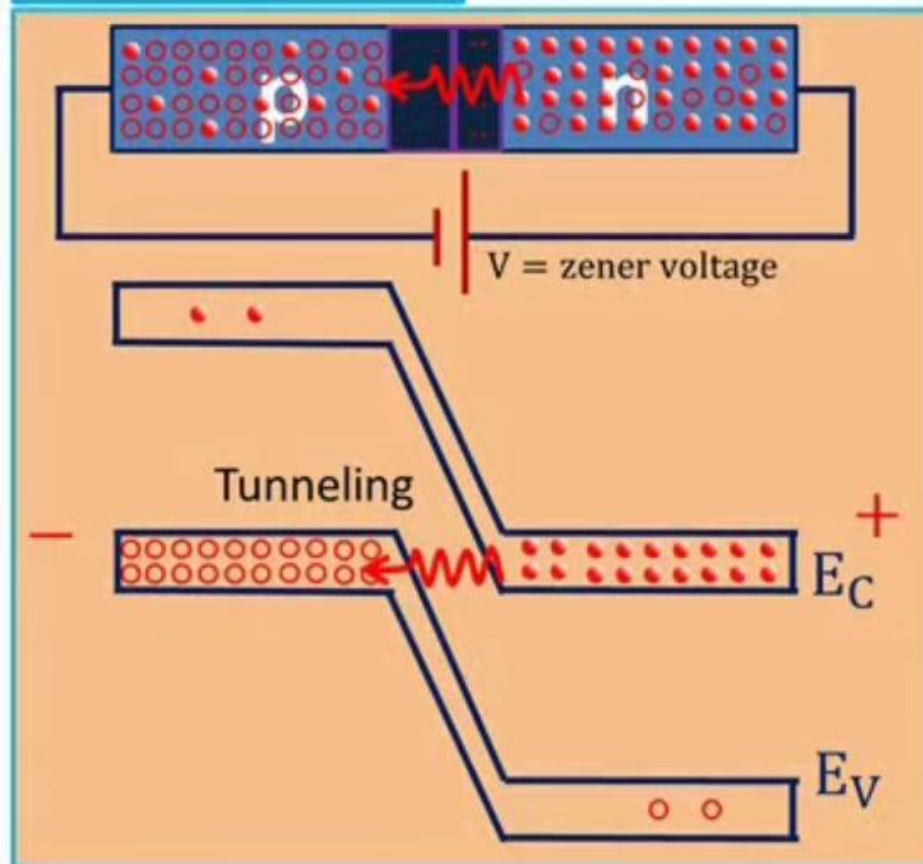
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Zener Diode

- **Zener Diode** is also called as “Breakdown Diode” 
- Same as the standard PN junction diode but are specially designed to have a **low pre-determined Reverse breakdown Voltage achieved by extremely high level of doping**
- The **zener diode** used in breakdown mode is the simplest types of **voltage regulator** and the point at which a zener diode breaks down or conducts is called the “Zener Voltage” (V_z)

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Zener Diode



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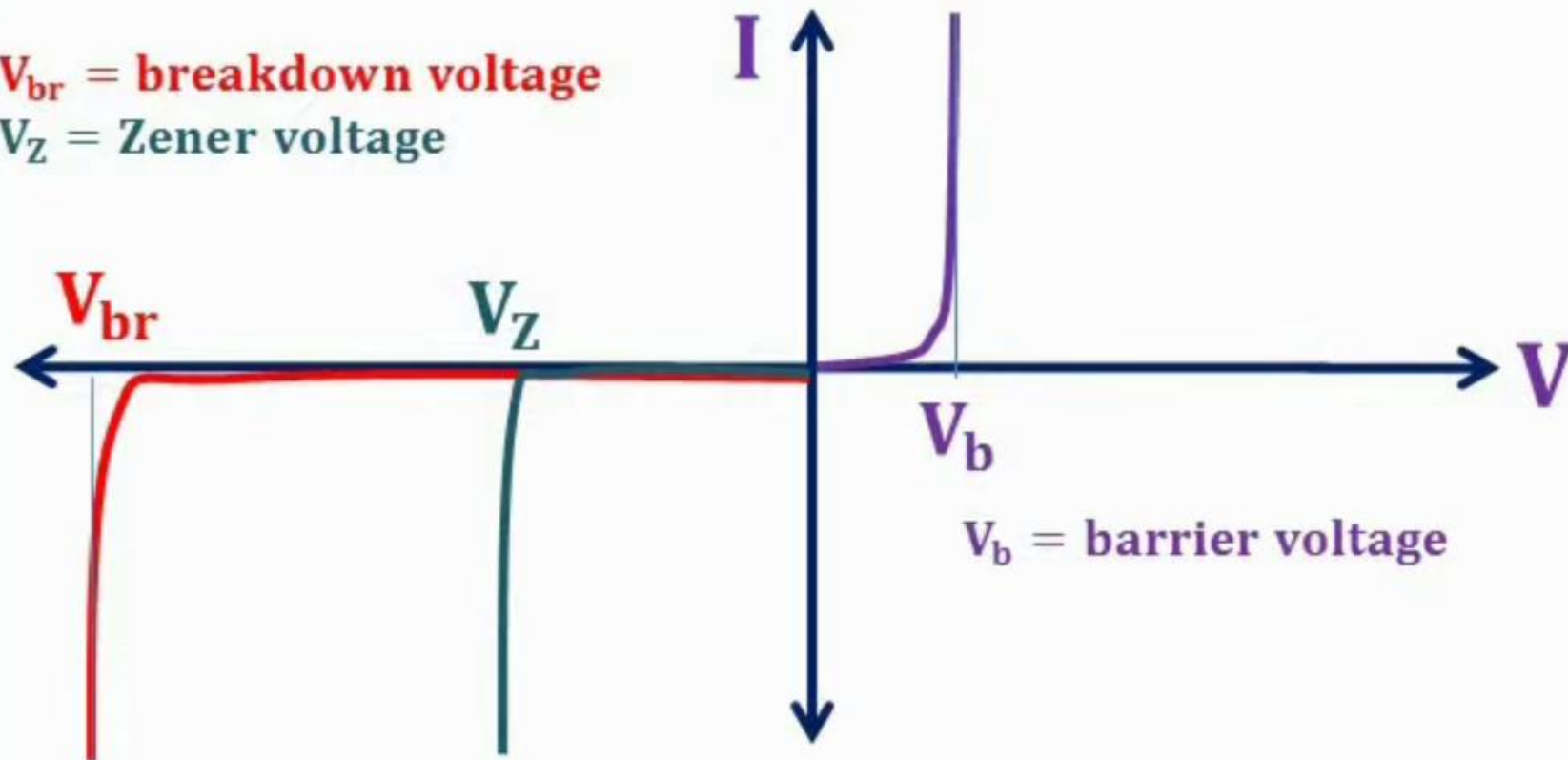
- The electric field becomes extremely strong, but at the same time, it is extremely narrow, hence many charge carriers can't get accelerated.
- Instead a quantum mechanical effect takes place. This is called as **quantum tunneling**.
- So there is no impact ionization, the electrons just tunnel through. (like digging a hole, instead of overcoming the mountain).
- **Zener breakdown**, only happens, when the doping concentration is extremely high. So that depletion region becomes very narrow.

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Zener Diode (V-I Characteristics)

V_{br} = breakdown voltage

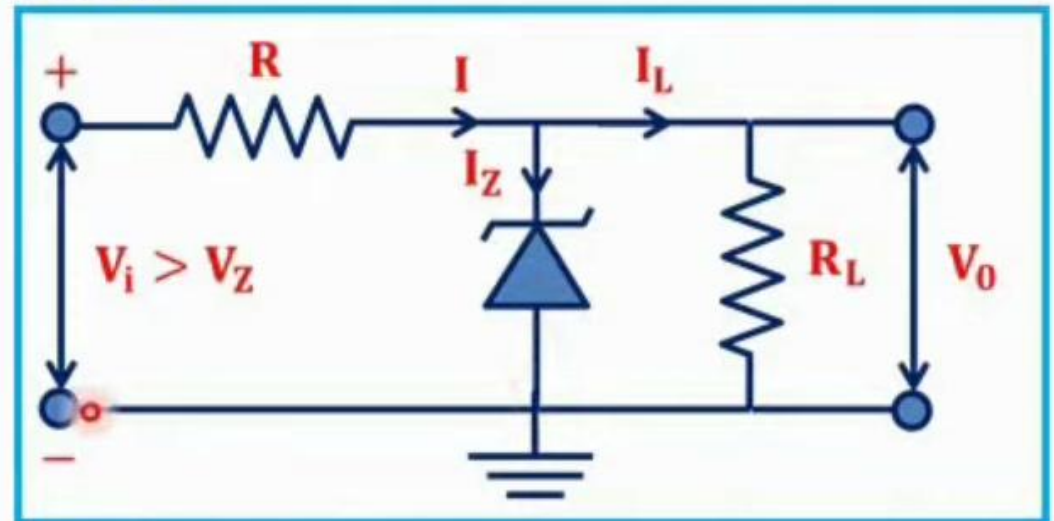
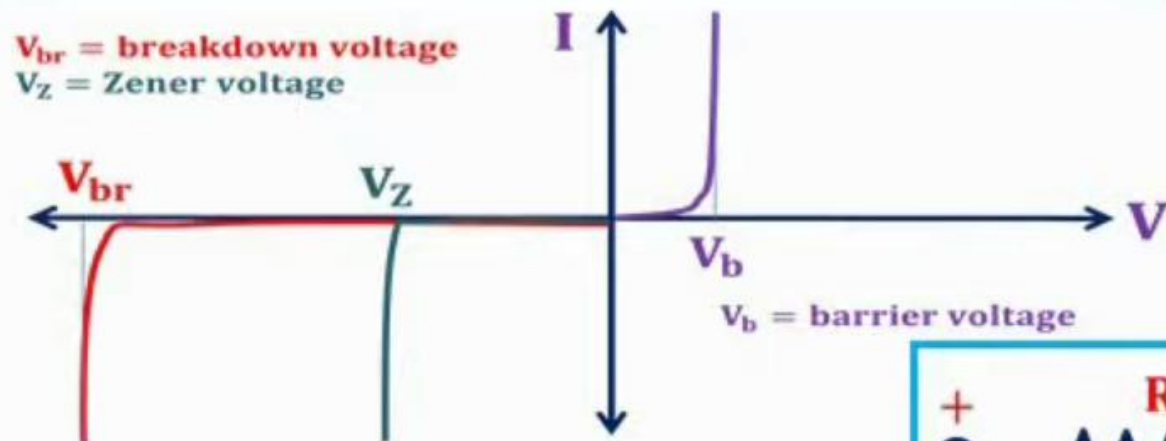
V_Z = Zener voltage



V_b = barrier voltage

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Zener Diode as a voltage regulator



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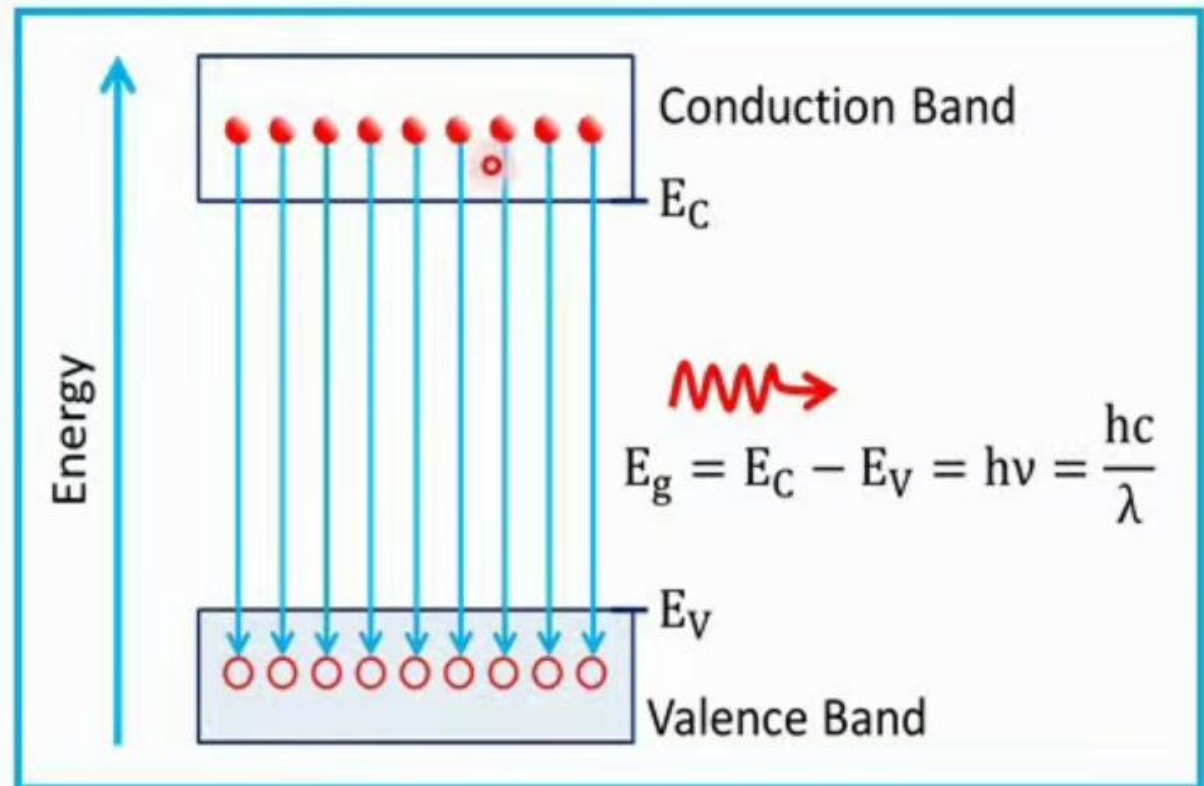
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AVALANCHE BREAKDOWN	ZENER BREAKDOWN
Avalanche breakdown happens due to avalanche effect. If the reverse bias voltage is very high, then the depletion region widens, and the electric field is quite strong. The minority charge carriers get accelerated in this depletion region, and gain kinetic energy. Once the field is strong enough, they can knock off, other electrons from the from the negative ions in the depletion region. This electron can in turn can knock off other electrons from the negative ions in the depletion region. So large number of electrons are available for conduction. However, due to Impact Ionization, heat is generated, and this heat could destroy the diode.	Zener breakdown, only happens, when the doping concentration is extremely high. So the depletion region becomes very narrow. The electric field becomes extremely strong, but at the same time, it is extremely narrow, hence many charge carriers can't get accelerated. Instead a quantum mechanical effect takes place. This is called as quantum tunneling. So there is no impact ionization, the electrons just tunnel through. (like digging a hole, instead of overcoming the mountain). So most electrons tunnel through, and some of-course will give impact ionization.

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Light Emitting Diode (LED)

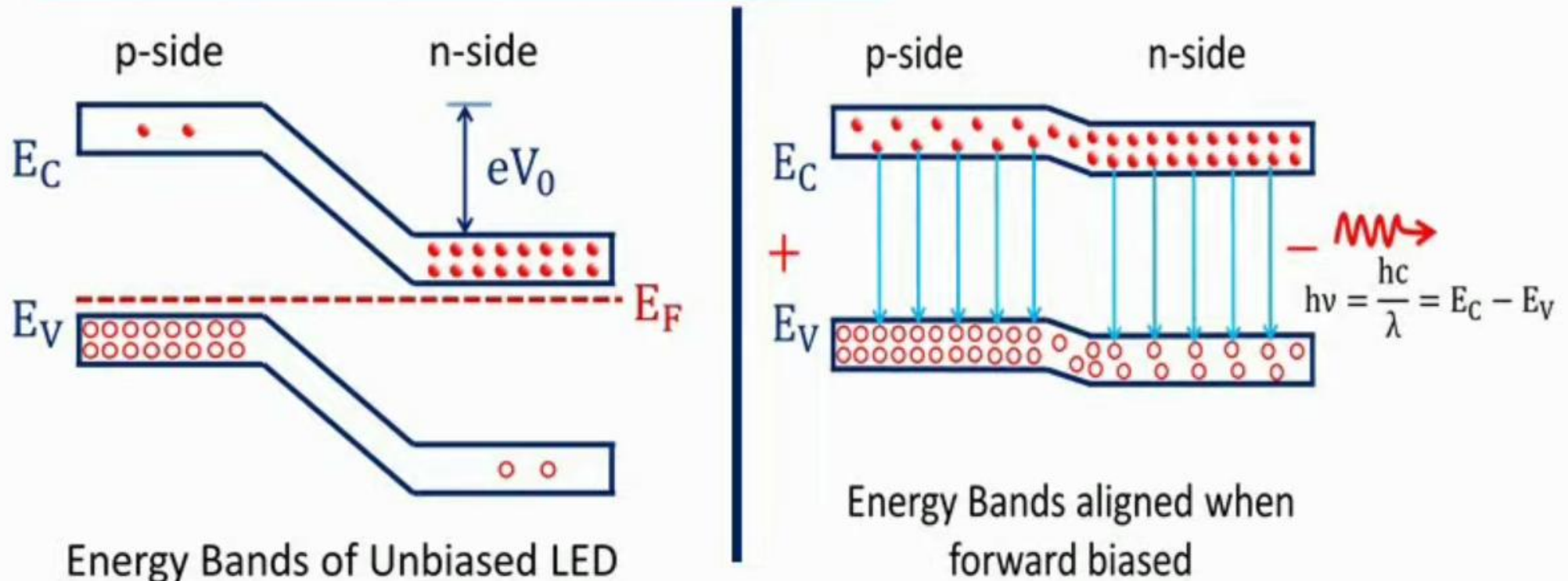
- LED is a diode which emits visible light when forward biased.
- LED works on the basic principle called 'recombination'.



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Energy Band diagram for LED

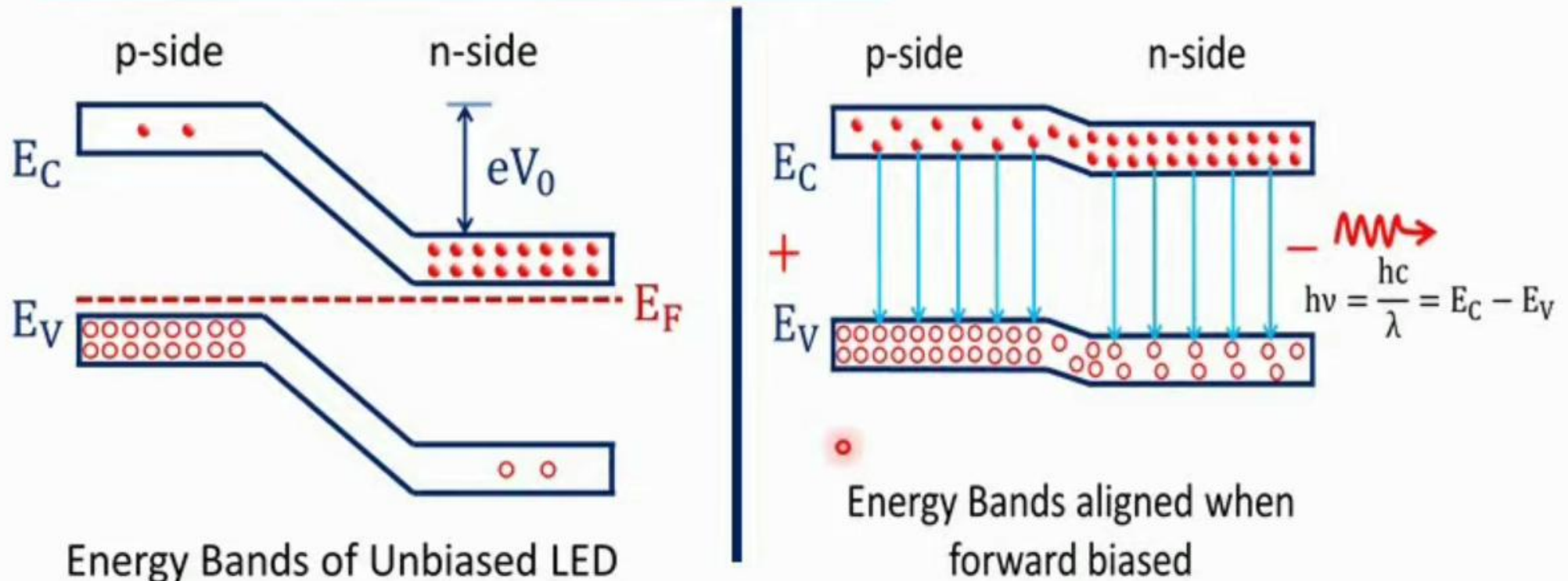
(Why LED is used in forward bias mode?)



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Energy Band diagram for LED

(Why LED is used in forward bias mode?)



Module 3: Semiconductor Physics

Light Emitting Diode (LED)

- Why Silicon and Germanium diodes can not be used to as LED?
 - These are **indirect gap** semiconductors
 - Energy Gap for Si = 1.1 eV
 - Energy Gap for Ge = 0.7 eV
 - Corresponding values of wavelength do not fall in visible region.
- Which materials are used for LED?
 - GaAs, GaAsP, GaP etc. which are obtained by mixing group 3 elements with group 5 elements.
 - These are **direct gap** semiconductors and the energy gap is such that corresponding wavelength falls in the visible region

$$E_g = E_C - E_V = h\nu = \frac{hc}{\lambda}$$

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Materials used in Light Emitting Diode (LED)

- Gallium arsenide (GaAs)
- Aluminium gallium arsenide (AlGaAs)
- Gallium arsenide phosphide (GaAsP)
- Aluminium gallium indium phosphide (AlGaInP)
- Gallium phosphide (GaP)
- Aluminium gallium phosphide (AlGaP)
- Aluminium nitride (AlN)
- Aluminium gallium nitride (AlGaN)

Module 3: Semiconductor Physics

Applications of LED



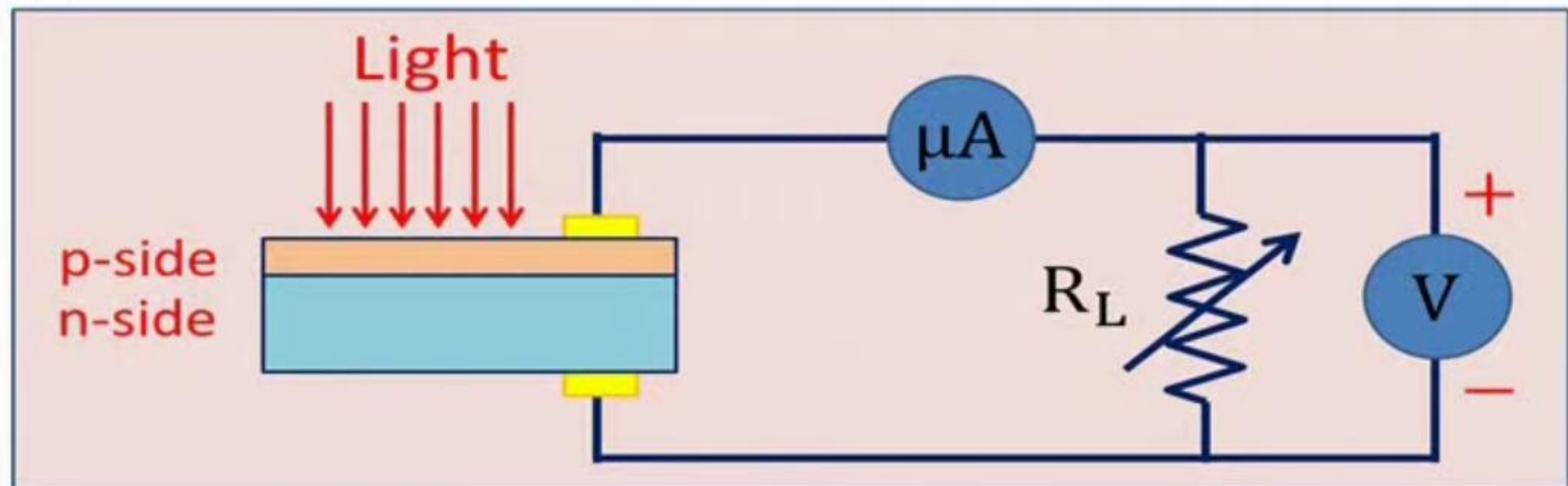
- Digital displays.
- Optical source in fibre optic communication.
- Infrared LEDs used in remote control devices.
- Widely used in automobile industry.

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Solar Cell (Photovoltaic Cell)

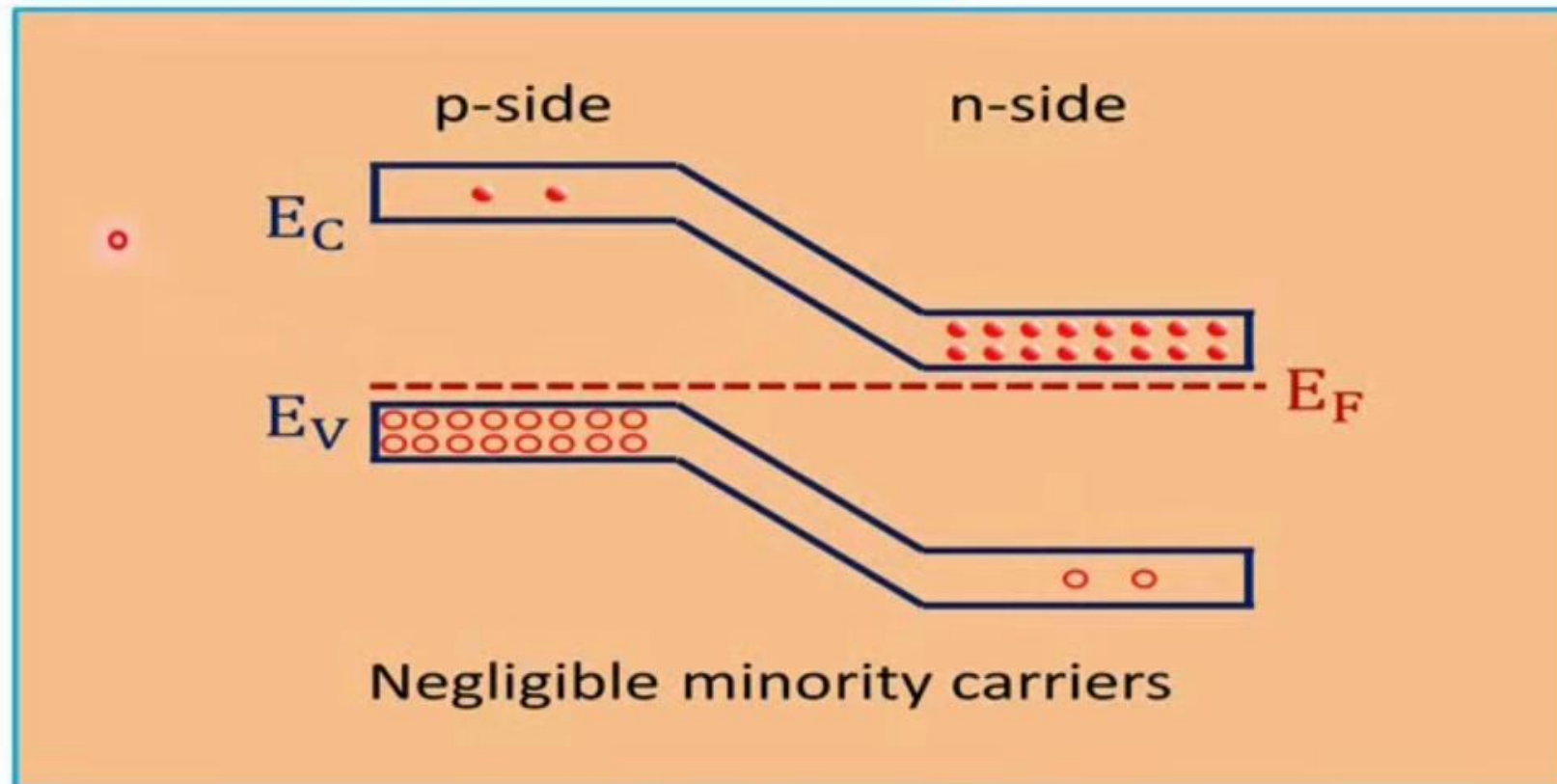
- It is alternative source of energy which converts solar energy into electrical energy.
- It is a p-n junction with a large surface area as compared to normal p-n junction diodes and very thin p-layer so that most of the light energy reaches the junction area.



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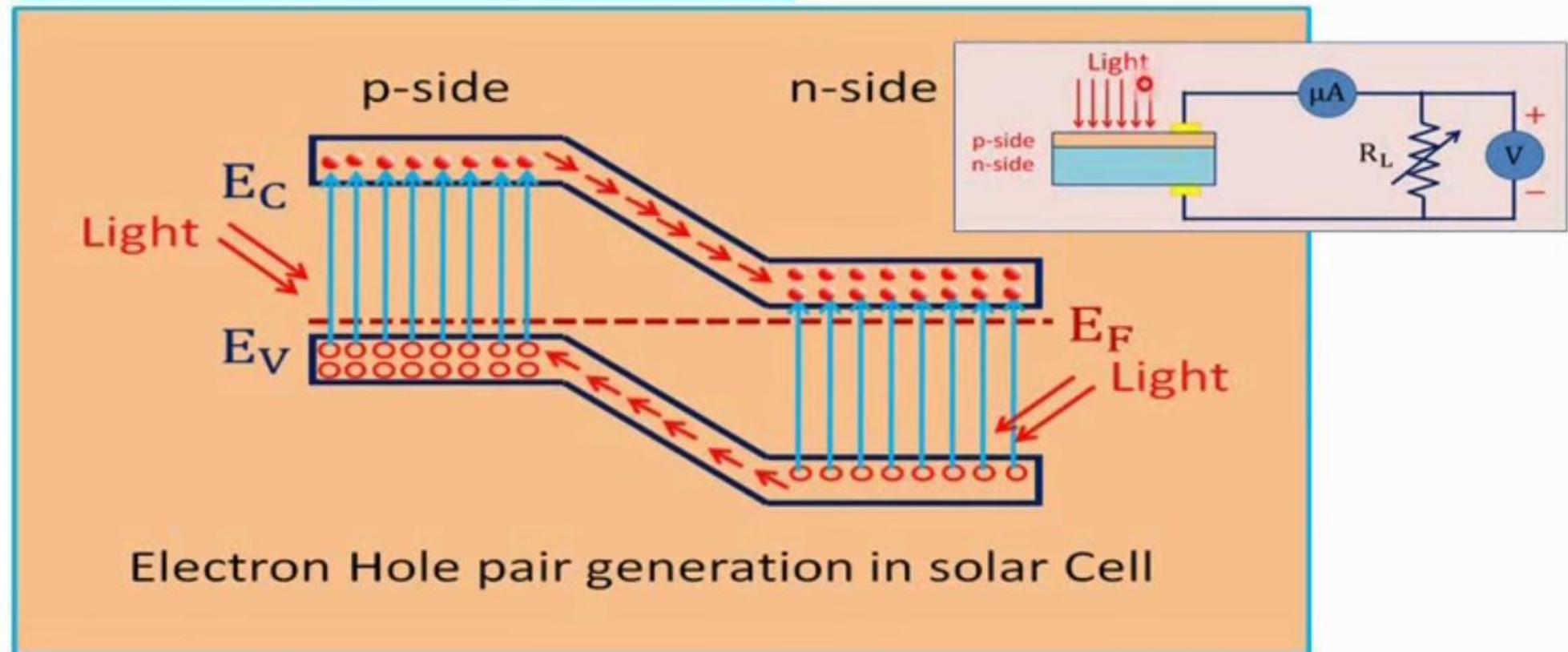
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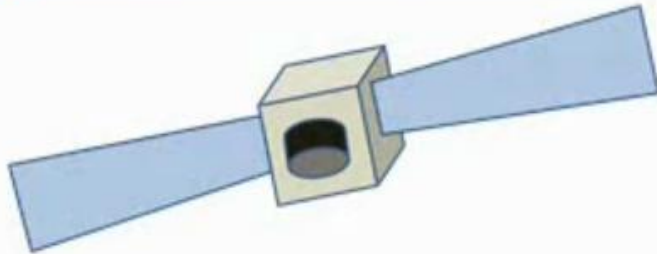
Solar Cell (Photovoltaic Cell)



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Applications of Solar Cell (Photovoltaic Cell)



- Used in satellites, space vehicles and at remote places as a source of energy.
- Used in battery charging system, outdoor lighting systems.
- Used in low power consumption electronic equipments like calculator.
- Pollution free source of energy.
- Maintenance free and cost effective in long run.

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Drawbacks of Using Solar Cell (Photovoltaic Cell)

- Initial cost is high
- Function is affected in rainy season when sufficient sunlight is not available.
- Requires a large space for high power operations.

